

NEURONET – AI-Driven Early Detection and Support System for Adolescent Mental Health.



**A Senior Project SRS Submitted for Partial Fulfillment of the Requirements
for BSc. Degree in Software Engineering**

**DEPARTMENT OF SOFTWARE ENGINEERING
COLLEGE OF COMPUTING AND INFORMATICS
HARAMAYA UNIVERSITY**

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HARAMAYA

DECLARATION SHEET

The **NEURONET – AI-Driven Early Detection and Support System for Adolescent Mental Health**. SRS outlined herein is the collective effort of the group members listed below, completed without external assistance, under the supervision of **prof. Gemechis Teshite**.

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DEFINITION, ABBREVIATIONS, AND ACRONYMS

This section defines the key terms, abbreviations, and acronyms used throughout this Software Requirements Specification (SRS) to promote clarity, consistency, and a shared understanding among all stakeholders. It ensures that technical terminology, system-specific concepts, and shortened forms are interpreted uniformly by readers including developers, designers, evaluators, and non-technical reviewers. By clearly explaining commonly used abbreviations and acronyms, this section helps reduce ambiguity, supports accurate communication, and improves the overall readability and effectiveness of the document, especially in areas involving system functionality, architecture, security, and ethical considerations.

Term / Acronym	Definition
NEURONET	The proposed AI-supported, non-clinical system designed to assist adolescents in emotional self-awareness through journaling, mood tracking, and consent-based support mechanisms.
AI (Artificial Intelligence)	Computational techniques that enable systems to perform tasks requiring human-like analysis, such as pattern recognition and sentiment assessment, without clinical interpretation.
NLP (Natural Language Processing)	A subfield of AI concerned with the analysis and interpretation of human language in textual form, used in NEURONET for high-level sentiment and trend analysis.
SRS (Software Requirements Specification)	A formal document that defines the functional and non-functional requirements of a software system, serving as an authoritative reference for development and evaluation.
RBAC (Role-Based Access Control)	A security principle in which system access and data visibility are determined by predefined user roles.
MVP (Minimum Viable Product)	The smallest set of system features required to demonstrate core functionality and meet project objectives within academic constraints.
Adolescent User	A primary system user aged 13–17 who interacts with NEURONET for journaling, mood tracking, and personal insight, under guardian-mediated consent.
Guardian	A parent or legal guardian who provides consent for adolescent participation and receives limited, aggregated system

	information as permitted.
Counselor	A verified professional user who provides educational content and scheduled or asynchronous support, without engaging in clinical diagnosis or emergency intervention.
Consent	Explicit authorization granted by the adolescent user (and guardian where applicable) that governs data sharing and alert visibility within the system.
Non-Clinical	A classification indicating that the system does not perform diagnosis, treatment, or professional medical decision-making.

Chapter One: Introduction

1.1 Document Scope

This Software Requirements Specification (SRS) document defines the scope, boundaries, and intended usage of the requirements specification for the proposed system, **NEURONET – AI-Driven Early Detection and Support System for Adolescent Mental Health**. The document establishes a shared understanding of what aspects of the system are specified, the level of detail provided, and the limitations within which the system is expected to operate. It serves as the authoritative reference for the system’s required behavior throughout the lifecycle of the senior capstone project.

This Software Requirements Specification (SRS) defines the conceptual and functional scope of **NEURONET**, a privacy-first digital support platform for adolescents. It specifies the system’s intended users, core functionalities, data handling and consent mechanisms, ethical constraints, and high-level operational environment.

The document focuses on the system’s externally observable behavior and responsibilities from a user and stakeholder perspective. While it identifies relevant platforms, environments, and constraints necessary for system operation, it intentionally avoids detailed design, algorithmic, and implementation-level specifications. The SRS therefore defines *what* the system must do and *under what conditions* it must operate, rather than *how* those requirements will be technically implemented.

The scope of this document explicitly includes the definition of system features related to user interaction, journaling and mood reflection, AI-assisted trend awareness, alert mechanisms, dashboards, and consent management. It also includes non-functional considerations such as privacy, security, usability, and ethical limitations, as they relate to the system’s overall quality and suitability for adolescent users. These elements are specified at a level appropriate for academic evaluation and for guiding subsequent design and development activities.

Conversely, this SRS intentionally excludes several aspects that fall outside its purpose or the project’s academic scope. Detailed system architecture, database schemas, algorithmic models, user interface designs, and technology-specific decisions are not covered in this document. Clinical assessment logic, medical treatment recommendations, and real-time crisis intervention mechanisms are also excluded, as NEURONET is explicitly defined as a non-clinical and non-diagnostic system. Any features or capabilities beyond the defined minimum viable product (MVP) are considered out of scope and are not addressed in this specification.

This document is organized into three main chapters to support clarity and traceability. Chapter One introduces the document itself, including its scope, purpose, conventions, intended audience, and organizational context. Chapter Two provides a high-level description of the NEURONET system, its users, operating environment, and key constraints. Chapter Three specifies the detailed functional and non-functional requirements that the implemented system must satisfy. This

structured organization allows readers to progressively move from contextual understanding to precise system expectations.

Different categories of readers are expected to use this document in different ways. Academic advisors and examiners may use the SRS to assess the project's problem definition, scope control, ethical awareness, and technical rigor. Student developers will rely on the SRS as a primary reference for understanding system expectations and for ensuring that development activities remain aligned with approved requirements. Reviewers and testers may use the document as a basis for validation, verification, and completeness checks. For all readers, this SRS represents the baseline against which the system's correctness and success will be evaluated.

This SRS also establishes a formal relationship with subsequent system documentation, particularly the Software Design Document (SDD). While the SRS defines *what* the system must do and the constraints under which it must operate, the SDD will describe *how* these requirements are realized through architecture, component design, data models, and algorithms. The SRS therefore acts as an input and contractual foundation for the SDD. Any design decisions documented in later artifacts must be traceable back to the requirements defined in this SRS to ensure consistency and academic integrity.

In summary, the scope of this document is to clearly and comprehensively define the boundaries, expectations, and intended use of the NEURONET system at the requirements level. By explicitly stating what is included, what is excluded, and how the document should be interpreted, this SRS provides a stable foundation for subsequent design, implementation, and evaluation activities within the senior capstone project.

1.2 Document Purpose

The purpose of this Software Requirements Specification (SRS) document is to formally define and communicate the expected behavior, boundaries, and constraints of the NEURONET system in a manner that is clear, precise, and suitable for academic evaluation. This document is created to serve as the primary requirements reference for the senior capstone project and to ensure that all stakeholders share a common understanding of the system's objectives and limitations before design and implementation activities begin.

A central problem addressed by this SRS is the risk of ambiguity and misinterpretation during software development. In the absence of a formal requirements specification, system development may rely on informal assumptions, evolving interpretations, or undocumented decisions, which can lead to inconsistencies, scope creep, and evaluation challenges. This document aims to eliminate such issues by explicitly stating the system's responsibilities, ethical boundaries, and expected interactions at an appropriate level of abstraction. By doing so, it provides a stable baseline against which future decisions can be assessed.

This SRS is also intended to guide the software development process in a structured and disciplined manner. For the development team, the document acts as a reference that informs architectural decisions, component design, and feature prioritization without dictating specific technologies or implementation approaches. By clearly defining system expectations, the SRS enables developers

to make informed design choices while remaining aligned with approved project objectives. Any subsequent design artifacts, including the Software Design Document (SDD), are expected to derive directly from the requirements defined in this specification.

In addition to guiding development, this document plays a critical role in system testing and validation. The requirements specified in this SRS provide the basis for defining test cases, validation scenarios, and acceptance criteria. Test activities can be systematically mapped to documented requirements, ensuring that system behavior can be objectively verified. This is particularly important for an academic project, where demonstrable correctness and traceability are key evaluation criteria.

The SRS also serves an important coordination function among stakeholders. Academic advisors use this document to assess the appropriateness of the project scope, the feasibility of the proposed system, and the students' understanding of ethical and technical constraints. Examiners rely on the SRS to evaluate whether the final system satisfies the originally approved requirements. For the student development team, the document provides a shared point of reference that supports consistent decision-making and accountability throughout the project lifecycle.

The importance of a formal requirements document is further amplified by the sensitive nature of the NEURONET system. As the system involves adolescent users and processes emotionally sensitive data, careful documentation is necessary to demonstrate responsible design intent, ethical awareness, and compliance with privacy expectations. This SRS makes explicit the system's non-clinical role, its privacy-first approach, and its consent-driven interactions, thereby reducing the risk of misuse or misinterpretation of the system's purpose.

In summary, the purpose of this SRS is to establish a clear, authoritative, and ethically grounded specification that guides the development, testing, and evaluation of NEURONET. By aligning stakeholders, reducing ambiguity, and providing a formal basis for assessment, this document contributes to both the technical rigor and academic integrity of the senior capstone project.

1.3 Document Conventions

This section defines the conventions adopted throughout this Software Requirements Specification (SRS) to ensure clarity, consistency, and traceability. The use of well-defined document conventions is essential in a formal academic specification, as it reduces ambiguity, supports systematic review, and enables different stakeholders to interpret the document in a uniform manner. These conventions are applied consistently across all chapters of this SRS.

1.3.1 Formatting Conventions

The document follows a structured, hierarchical formatting style commonly used in academic and industry-standard SRS documents. Chapters, sections, and subsections are numbered using a decimal notation (e.g., 1, 1.1, 1.1.1) to clearly indicate logical relationships and allow precise referencing. Headings and subheadings are used to separate conceptual topics and improve readability.

Text is written in formal academic language, avoiding informal expressions, personal opinions, or persuasive wording. Emphasis is applied sparingly using bold formatting for key terms, section titles, and requirement identifiers. Lists are used where appropriate to present related items clearly, while paragraphs are employed for explanatory and descriptive content. These formatting choices support ease of navigation for reviewers and examiners.

1.3.2 Requirement Statement Style

Although detailed requirements are specified later in the document, a consistent requirement statement style is adopted throughout the SRS. The keyword “**shall**” is used exclusively to indicate mandatory system behavior or constraints. This usage aligns with internationally recognized SRS standards and removes uncertainty regarding whether a statement is optional or compulsory.

The keywords “**should**”, “**may**”, or “**could**” are deliberately avoided in requirement definitions to prevent ambiguity. Where descriptive or explanatory statements are necessary, they are clearly distinguished from requirements through contextual wording and placement outside requirement lists. This convention ensures that only explicitly labeled statements are interpreted as binding requirements.

1.3.3 Naming Conventions

Consistent naming conventions are applied to system entities, user roles, and document elements. System components and features are referred to using descriptive and unambiguous names. User roles such as *adolescent user*, *guardian*, *counselor*, and *system administrator* are consistently capitalized and used uniformly throughout the document.

Requirement identifiers follow a standardized naming format to support traceability. Functional requirements are labeled as **FR-XX**, while non-functional requirements are labeled as **NFR-XX**, where *XX* represents a sequential number. These identifiers are unique and stable, allowing requirements to be referenced across testing documents, design artifacts, and evaluation reports.

1.3.4 Diagram and Notation Conventions

Where diagrams are referenced or included in supporting documentation, standard software engineering notation is assumed. Unified Modeling Language (UML) conventions are used for diagrams such as use case diagrams, sequence diagrams, and class diagrams, where applicable. Each diagram is expected to include a clear title, legend (if necessary), and descriptive labels to avoid misinterpretation.

The use of standardized diagram notation ensures that visual representations of the system can be easily understood by readers with a software engineering background and can be consistently mapped to the textual requirements defined in this SRS.

1.3.5 Terminology Consistency

Terminology is used consistently throughout the document to avoid confusion. Key concepts are introduced once and reused with the same meaning in all subsequent sections. For example, the terms *alert*, *self-alert*, *trusted-adult alert*, and *emergency alert* are applied consistently to reflect the system’s tiered notification model.

When multiple terms could describe the same concept, a single term is selected and used uniformly. This practice minimizes misinterpretation and ensures that readers can reliably associate specific terms with specific system concepts.

1.3.6 Acronyms and Abbreviations

Acronyms and abbreviations are used sparingly and only where they improve readability. When an acronym is first introduced, it is written in full followed by the abbreviated form in parentheses. For example, *Software Requirements Specification (SRS)* and *Artificial Intelligence (AI)*. After initial definition, the abbreviated form may be used consistently throughout the document.

Uncommon or project-specific abbreviations are avoided unless they are clearly defined at their first occurrence and listed in this section. This approach ensures that the document remains accessible to academic reviewers and stakeholders who may not be familiar with specialized technical terminology or shorthand.

1.3.7 Rationale for Using Document Conventions

The conventions defined in this section are intended to improve clarity, reduce ambiguity, and support traceability across the project lifecycle. Clear formatting and consistent terminology allow readers to quickly locate information and understand relationships between sections. Standardized requirement statements and identifiers enable systematic testing, verification, and evaluation.

In an academic setting, particularly for a sensitive system involving adolescent users and personal data, these conventions contribute to transparency and accountability. They ensure that system expectations are clearly documented, ethically framed, and objectively assessable, thereby enhancing the overall quality and credibility of the SRS.

1.4 Intended Audience

This Software Requirements Specification (SRS) is intended to be used by multiple audiences with distinct roles and responsibilities throughout the lifecycle of the NEURONET project. Each audience interacts with the document from a different perspective; therefore, this section clarifies their expectations, how they are expected to use the SRS, and which parts of the document are most relevant to their needs. Defining the intended audience explicitly helps ensure that the document fulfills its role as a shared and authoritative reference.

1.4.1 Software Developers

The primary audience for this SRS is the team of software engineering students responsible for designing, implementing, and testing the NEURONET system. For developers, this document serves as the definitive description of system behavior, boundaries, and constraints. It establishes a clear understanding of what the system is expected to do, thereby reducing ambiguity during development and preventing scope creep.

Developers are expected to use the SRS as a baseline when making design decisions, implementing features, and verifying system functionality. Chapters Two and Three are particularly relevant to this audience, as they describe the system context, user classes, and detailed functional requirements. By adhering strictly to the requirements defined in this document, developers can ensure that the implemented system aligns with academic expectations and ethical constraints without relying on informal interpretations or assumptions.

1.4.2 Project Advisor

The project advisor uses this SRS as a supervisory and evaluative tool throughout the duration of the capstone project. From the advisor's perspective, the document provides evidence that the project scope is well-defined, academically appropriate, and feasible within the constraints of a senior-level software engineering course.

The advisor is expected to review the SRS to assess the clarity, completeness, and consistency of the proposed system requirements. Chapter One is especially relevant for understanding the project context, purpose, and assumptions, while Chapter Three allows the advisor to evaluate whether the functional scope is realistic and aligned with ethical and academic standards. The SRS also enables the advisor to provide structured feedback and guide the team toward disciplined engineering practices.

1.4.3 Academic Examiners

Academic examiners and evaluation committee members represent another critical audience for this document. For examiners, the SRS serves as an objective reference against which the success and completeness of the implemented system can be assessed. It defines the criteria used to judge whether the project meets its stated objectives and whether the delivered system satisfies the approved requirements.

Examiners are likely to focus on the overall coherence of the document, the justification of the system scope, and the precision of the requirements. Chapter One provides essential context for understanding the problem domain and the rationale behind the system boundaries, while Chapter Three is central to evaluating whether the implementation corresponds to the documented requirements. A well-structured SRS enables examiners to conduct fair and transparent evaluation without relying on undocumented expectations.

1.4.4 Future Maintainers and Extenders

Although NEURONET is developed as a senior capstone project, the SRS is also intended to support potential future maintainers, researchers, or student teams who may extend or adapt the system. For this audience, the document acts as a knowledge transfer artifact that explains the original intent, scope, and limitations of the system.

Future maintainers are expected to use the SRS to understand what functionality already exists, which design decisions were constrained by ethical or academic considerations, and where extensions could be introduced without violating the original system boundaries. Chapters Two and Three are particularly valuable in this context, as they describe the system structure and behavior independently of any specific implementation technology. This long-term perspective enhances the sustainability and reusability of the project.

1.4.5 Summary of Audience Alignment

By explicitly addressing the needs of developers, advisors, examiners, and future maintainers, this SRS functions as a shared point of reference across all stakeholders. Each audience can extract the information relevant to their role while relying on a single, consistent specification. This alignment reduces miscommunication, supports objective evaluation, and reinforces disciplined software engineering practices throughout the NEURONET project lifecycle.

1.5 Team Organization

The NEURONET project is developed as a senior capstone project by a multidisciplinary team of undergraduate students from the Department of Software Engineering at Haramaya University. The team organization is designed to support effective collaboration, clear accountability, and systematic development of both the system and its accompanying documentation. Defining roles and responsibilities at this stage ensures that project activities are coordinated and aligned with the objectives outlined in this Software Requirements Specification (SRS).

1.5.1 Team Structure

The project team follows a structured yet collaborative organization model appropriate for an academic software engineering project. While each team member is assigned a primary role, the team operates with shared responsibility for overall project success. This structure encourages peer learning, knowledge sharing, and collective problem-solving, while still maintaining clarity regarding ownership of specific tasks.

The team operates under the academic supervision of a project advisor, who provides guidance, feedback, and oversight. Decision-making within the team is conducted through regular discussions and consensus, with the team leader responsible for coordination and final alignment with project goals.

1.5.2 Roles and Responsibilities

Each role within the team is defined to ensure balanced workload distribution and effective coverage of technical and documentation tasks:

The project team consists of the following members,

each assigned specific roles:

- Garoma Makure: Team Lead and Developer
- Hamdi Abdulfetah: Frontend Developer
- Firafis Berhanu: Backend Developer
- Muaz Shewabu: UI/UX Designer

Although responsibilities are differentiated, team members are expected to collaborate across roles, particularly during requirement clarification, design reviews, and testing activities.

1.5.3 Collaboration and Communication Approach

The team adopts a collaborative development approach supported by regular meetings, shared documentation tools, and version control practices. Progress is reviewed periodically to ensure alignment with the approved SRS and to identify potential risks or misunderstandings early. Communication with the project advisor is conducted at defined milestones, particularly after the completion of major documentation deliverables and development phases.

Peer reviews are encouraged for both code and documentation artifacts. This practice supports quality improvement, reinforces shared understanding of system requirements, and helps maintain consistency across different components of the project.

1.5.4 Alignment with SRS Development

Responsibilities within the team are explicitly aligned with the development and maintenance of this SRS. All team members contribute to requirement validation by reviewing the document to ensure accuracy, completeness, and feasibility. The documentation and quality assurance role ensures that updates to the SRS reflect agreed-upon changes and remain consistent with academic standards.

By maintaining a clear organizational structure and well-defined responsibilities, the team ensures that the SRS functions as a living reference throughout the project lifecycle. This alignment supports disciplined engineering practices and strengthens the traceability between documented requirements and the final system implementation.

CHAPTER TWO: SYSTEM DESCRIPTION

2.1 System Overview

2.1.1 Introduction to the System

NEURONET is an Artificial Intelligence (AI)–assisted early awareness and support system designed to promote mental well-being among adolescents through structured digital self-expression, ethical AI-based analysis, and controlled human support mechanisms. The system is implemented as both a mobile application and a web-based platform to accommodate different user roles, including adolescents, guardians, counselors, and system administrators.

The system is designed around the principle of **ethical, non-intrusive support**. Adolescents use NEURONET primarily as a digital journaling and reflection platform, where they can record thoughts, emotions, and daily experiences in a secure environment. These inputs are analyzed in the background by AI models to identify emotional patterns and trends over time. Importantly, the AI does not provide therapy, diagnosis, or personalized advice.

NEURONET introduces a **role-based interaction model** in which guardians, counselors, and adolescents interact through well-defined and privacy-respecting boundaries. Guardians oversee registration and consent, counselors provide professional support when required, and AI functions strictly as an analytical and informational component.

2.1.2 Objectives of the System

The primary objectives of the NEURONET system are:

- To provide adolescents with a safe digital space for emotional expression through journaling and reflection.
- To utilize AI-based sentiment analysis to detect emotional patterns and behavioral trends without performing diagnosis.
- To enable adolescents to interact with an AI assistant for **system-related guidance and general mental health information**, while explicitly restricting advice, therapy, or judgment.
- To support controlled, consent-based communication between adolescents and assigned counselors when early warning indicators are detected.
- To enable guardians to manage adolescent registration, consent, and receive summarized alerts without accessing private journal content.
- To facilitate professional oversight by counselors through dashboards, alerts, and structured communication tools.

2.1.3 Problem Statement Summary

Adolescent mental health challenges are increasingly prevalent, yet many early warning signs remain unnoticed due to stigma, lack of awareness, or limited access to professional support.

Existing digital solutions often either rely on self-assessment tools with limited continuity or introduce AI-driven advice systems that raise ethical and reliability concerns.

NEURONET addresses this problem by combining **continuous emotional self-expression**, **background AI analysis**, and **human-centered support**. The system avoids intrusive monitoring and unsafe AI interaction by focusing on trend detection rather than immediate judgment, ensuring that professional counselors remain central to meaningful intervention.

2.1.4 Significance of the System

The significance of NEURONET lies in its balanced integration of technology, ethics, and human support. By positioning AI as an analytical tool rather than a decision-maker, the system reduces the risk of misinformation while still benefiting from data-driven insights.

The system empowers adolescents to engage in self-reflection without fear of surveillance, enables guardians to remain informed through alerts and summaries, and allows counselors to focus their attention where it is most needed. This preventive, awareness-oriented approach contributes to early support-seeking behavior and improved mental health literacy.

2.1.5 Technologies and Tools Overview

NEURONET employs modern, scalable technologies to support secure interaction and intelligent processing, including:

- Cross-platform mobile and web application frameworks
- Backend services for authentication, role management, and system logic
- AI and Natural Language Processing (NLP) libraries for sentiment analysis
- Secure relational and document-based databases for structured and unstructured data
- Cloud-based infrastructure to ensure availability, scalability, and data protection

All technologies are selected to support maintainability, ethical AI usage, and role-based security enforcement.

2.1.6 Overall System Workflow

The high-level workflow of the NEURONET system is as follows:

1. A guardian registers an adolescent by providing required personal and contact information.
2. The adolescent activates the account by setting a password and accessing the system using guardian-provided credentials.
3. The adolescent uses the system to write journal entries and record mood reflections.
4. AI modules analyze textual data in the background to generate sentiment scores and detect emotional trends.
5. When predefined thresholds are met, alerts are generated and displayed on counselor and guardian dashboards.

6. Counselors review alerts and trends for assigned adolescents and may initiate consent-based communication.
7. Adolescents may interact with counselors privately after guardian approval.
8. Throughout the process, consent rules, privacy controls, and access limitations are strictly enforced.

2.2 System Scope

2.2.1 Purpose of the System

The purpose of NEURONET is to function as a **supportive, awareness-focused digital platform** for adolescent mental well-being. The system is intended to assist in identifying emotional trends early and facilitating appropriate human support rather than replacing professional mental health services.

NEURONET is explicitly designed as a **non-clinical system**. It does not provide medical diagnosis, therapy, or treatment recommendations. Instead, it complements existing support structures by enabling early awareness and structured communication.

2.2.2 In-Scope Functionalities

The following functionalities are included within the system scope:

- Guardian-led registration of adolescent users
- Secure authentication and role-based access control
- Adolescent journaling and mood tracking features
- AI-based sentiment analysis and emotional trend detection
- AI informational interaction limited to system usage and general well-being information
- Alert generation based on predefined emotional patterns
- The system shall allow adolescents to communicate with counselors only after guardian approval, primarily through scheduled sessions or asynchronous messaging (not instant chat by default). Private, consent-based chat between adolescents and counselors
- Guardian–counselor communication for coordination purposes
- Counselor-managed informational channels that adolescents can follow, react to, and comment on
- Dashboards for guardians and counselors displaying summarized insights
- Consent management, privacy enforcement, and audit logging

2.2.3 Out-of-Scope Functionalities

The following functionalities are explicitly excluded from the system:

- Medical diagnosis or clinical decision-making
- AI-provided therapy, advice, or judgment
- Emergency or crisis intervention services

- Real-time audio or video communication
- Direct guardian access to adolescent journal content
- Offline monitoring or wearable device integration
- Integration with hospital or electronic health record systems

2.2.4 System Boundaries

NEURONET operates strictly within its defined digital boundaries. The system processes only user-generated data created within the platform and does not ingest external data sources. AI analysis results are used solely for awareness and alert generation and must not be interpreted as medical conclusions.

All interactions between roles are governed by consent, role permissions, and privacy constraints to ensure ethical operation and data protection.

2.3 User Classes and Characteristics

This section describes the different categories of users who interact with the NEURONET system. Each user class has distinct roles, access privileges, responsibilities, and expectations. Clearly defining user characteristics is essential for ensuring system usability, security, ethical compliance, and effective role-based access control. The identified user classes directly influence system functionality and interaction flows.

2.3.1 Adolescent User

The Adolescent User is the primary user of the NEURONET system. Adolescents interact mainly through the mobile application to record journal entries, reflect on emotions, and engage with awareness-oriented system features. Adolescents do not register themselves independently; instead, their accounts are created by guardians to ensure proper consent and oversight.

Characteristics

- Aged between **13 and 17 years**
- Varying levels of emotional maturity and self-expression ability
- Basic to moderate digital literacy
- Highly sensitive to privacy, trust, and autonomy
- May be hesitant to seek help through traditional means

Needs and Expectations

- A simple, non-judgmental, and emotionally safe user interface
- Assurance that private journal entries are not directly accessible to guardians
- Clear understanding of how AI analyzes content and what its limitations are
- Ability to interact with an AI assistant for **system-related guidance and general information only**

- Option to communicate privately with an assigned counselor after guardian approval
- Access to personal emotional trends and summaries without diagnostic labels

The system must prioritize confidentiality, transparency, and emotional safety for adolescent users, as their engagement is fundamental to the effectiveness of the platform.

2.3.2 Guardian

Guardians are parents or legal caretakers responsible for adolescents using the NEURONET system. Guardians play a critical role in registration, consent management, and high-level monitoring, while respecting adolescent privacy.

Characteristics

- Varying levels of technical proficiency
- Strong concern for adolescent safety and well-being
- Prefer concise and clearly presented information
- Responsible for initial account creation and consent approval

Needs and Expectations

- Ability to register adolescents and provide required personal information
- Access to **summarized insights, trends, and alerts**, not raw journal data
- Clear notifications when emotional patterns reach predefined thresholds
- Communication channel with counselors when coordination is required
- Assurance that adolescent autonomy and privacy are preserved

Guardians are intentionally restricted from accessing private journals and counselor–adolescent conversations to maintain trust and ethical boundaries.

2.3.3 Counselor / Mental Health Professional

Counselors are trained mental health professionals who provide human support and professional oversight. Each counselor may be assigned by the system to manage and monitor multiple adolescents (e.g., up to 20) based on availability and workload balancing.

Characteristics

- Professionally trained in mental health or counseling
- Experienced in interpreting emotional trends and behavioral indicators
- Operate under ethical and professional standards
- Use the system as a supplementary awareness tool

Needs and Expectations

- Access to dashboards displaying aggregated trends and alerts for assigned adolescents
- Clear separation between AI-generated indicators and clinical judgment
- Ability to initiate **consent-based private communication** with adolescents
- Ability to communicate with guardians when necessary without exposing private adolescent content
- Ability to create and manage informational channels where adolescents can follow, react, and comment

The system does not provide counselors with diagnostic authority; it supports awareness, monitoring, and professional follow-up.

2.3.4 System Administrator

The System Administrator manages the technical and operational aspects of the NEURONET system and does not participate in mental health interpretation or user interactions.

Characteristics

- High technical expertise in system administration
- Authorized to manage system configurations and user roles

Needs and Expectations

- Tools for managing user accounts, roles, and permissions
- Ability to configure alert thresholds and counselor assignments
- Access to system logs and performance monitoring
- Responsibility for ensuring security, availability, and compliance

2.3.5 Summary of User Class Interactions

Each user class interacts with the NEURONET system under strictly enforced role-based access controls. Adolescents focus on self-expression and awareness, guardians oversee consent and receive alerts, counselors provide professional support, and administrators maintain system integrity. This separation of responsibilities ensures ethical data usage, privacy protection, and secure system operation.

2.4 Business Rules

Business rules define the policies, constraints, and standards that govern the behavior of the NEURONET system. These rules ensure ethical AI usage, data protection, regulatory compliance, and consistent system operation. They directly impact both functional and non-functional requirements.

2.4.1 Ethical and Privacy Rules

Rule ID	Business Rule Description	Impacted Area
BR-01	The system shall not provide medical diagnosis, therapy, or treatment recommendations.	Ethical, Functional
BR-02	The system shall operate strictly as a non-clinical awareness and support platform.	System Scope
BR-03	Users shall be clearly informed of the system's purpose, capabilities, and limitations during onboarding.	Usability
BR-04	Adolescent journal content shall not be directly accessible to guardians.	Privacy

Table 1 Ethical and Privacy Rules

2.4.2 Data Access and Consent Rules

Rule ID	Business Rule Description	Impacted Area
BR-05	Adolescent accounts shall be registered only by guardians.	Registration
BR-06	Adolescent data shall be shared with counselors or guardians only after explicit consent approval.	Privacy, Security
BR-07	Adolescents shall have the right to revoke previously granted consent at any time.	Functional
BR-08	Counselor–adolescent communication shall require guardian approval.	Access Control

Table 2 Data Access and Consent Rules

2.4.3 Security and Data Protection Rules

Rule ID	Business Rule Description	Impacted Area
BR-09	All sensitive data shall be encrypted during transmission and storage.	Security
BR-10	All users shall be authenticated before accessing system resources.	Authentication
BR-11	Role-based authorization shall restrict access to permitted data and functions only.	Authorization
BR-12	System activity logs shall be maintained for monitoring and auditing purposes.	Maintenance

Table 3 Security and Data Protection Rules

2.4.4 AI Usage and Output Rules

Rule ID	Business Rule Description	Impacted Area
BR-13	AI shall analyze user-generated content only for sentiment and trend detection.	Functional
BR-14	AI shall provide informational responses about system usage and general well-being only.	Usability
BR-15	AI shall refuse requests for advice, therapy, diagnosis, or judgment and redirect users to counselors.	Ethical
BR-16	AI outputs shall be presented as indicators or trends, not definitive conclusions.	Functional

Table 4 AI Usage and Output Rules

2.4.5 Operational and Compliance Rules

Rule ID	Business Rule Description	Impacted Area
BR-17	Counselors may be assigned to multiple adolescents based on workload balancing.	System Logic
BR-18	Counselor-guardian communication shall not expose private adolescent journal data.	Privacy
BR-19	The system shall comply with institutional, academic, and ethical research guidelines.	Compliance

Table 5 Operational and Compliance Rule

2.5 Operating Environment

This section describes the environment in which the NEURONET system will operate after development and deployment. It outlines the required hardware, software, network, and infrastructure components necessary to support secure interaction, AI-assisted analysis, and role-based access. Defining the operating environment ensures compatibility, reliability, scalability, and maintainability of the system.

2.5.1 Hardware Environment

The NEURONET system is designed to operate on commonly available consumer hardware to ensure accessibility across all user roles without requiring specialized equipment.

Client Devices

- **Adolescent Users:**
 - Android smartphones for daily journaling, AI interaction, and counselor communication
- **Guardians**
 - Android smartphones for guardian oversight, viewing adolescent mood dashboards, managing consent, and communicating with counselors.
- **Counselors, and Administrators:**
 - Desktop or laptop computers for dashboard access and system management

Minimum Hardware Requirements

- **Processor:** Dual-core processor or higher
- **Memory:**
 - Minimum 2 GB RAM for mobile devices
 - Minimum 4 GB RAM for desktop or laptop systems
- **Storage:** At least 200 MB of free storage for application data and caching

The system does not rely on device-specific sensors or peripherals, ensuring wide usability and ease of deployment.

2.5.2 Software Environment

The software environment specifies the operating systems and browsers required to access NEURONET services effectively.

Operating Systems

- Android OS version 8.0 or higher
- Microsoft Windows 10 or higher
- Linux distributions (Ubuntu and equivalent)
- macOS

Web Browsers

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

All supported browsers must comply with modern web standards, including HTML5, CSS3, and JavaScript, to ensure consistent user experience across platforms.

2.5.3 Application and Development Environment

The NEURONET system is implemented using modern, scalable, and widely supported technologies suitable for academic and production-level development.

Component	Technology
Mobile Application	Flutter
Web Application	Next.js
Backend Framework	Django / FastAPI
Programming Languages	Python, JavaScript, Dart
Database Systems	PostgreSQL, MongoDB

Component	Technology
AI / NLP Tools	Open-source Python NLP libraries
API Communication	RESTful APIs

Table 6 Application and Development Environment

These technologies are selected for their reliability, scalability, community support, and suitability for implementing ethical AI and role-based access control.

2.5.4 Server and Hosting Environment

The NEURONET system is hosted on a cloud-based infrastructure to ensure availability, scalability, and secure data handling.

- Cloud servers with scalable computing resources
- Secure storage for structured and unstructured data
- Automated backup and recovery mechanisms
- Load balancing to support multiple concurrent users
- Centralized logging and system monitoring

The hosting environment supports near real-time AI processing and dashboard updates while maintaining system stability.

2.5.5 Network and Communication Environment

The system requires stable network connectivity to support secure communication between clients and backend services.

- Reliable internet connectivity for all users
- Secure communication using HTTPS protocols
- Encrypted data transmission to prevent unauthorized interception
- REST-based communication between frontend and backend services

Network security measures ensure confidentiality and integrity of sensitive adolescent data.

2.6 Design and Implementation Constraints

This section identifies constraints that limit design and implementation choices for the NEURONET system. These constraints arise from academic requirements, ethical considerations, technical limitations, and operational conditions.

2.6.1 Academic Constraints

As a senior capstone project, NEURONET is subject to academic and institutional limitations, including:

- Fixed development timeline defined by the academic calendar
- Limited team size and development resources
- Mandatory adherence to departmental SRS and SDD documentation standards
- Requirement to maintain a feasible and well-defined project scope

These constraints influence feature selection, implementation depth, and delivery schedule.

2.6.2 Ethical and Legal Constraints

Given the system’s focus on adolescents and sensitive mental health–related data, ethical and legal considerations strongly influence system design.

- Mandatory guardian involvement in adolescent registration
- Consent-based data sharing and communication
- Strict prohibition of medical diagnosis, therapy, or AI-generated advice
- Protection of adolescent privacy, including restricted guardian access to private journals
- Responsible and transparent use of AI-generated indicators

These constraints ensure ethical operation and user trust.

2.6.3 Technical Constraints

Technical limitations affect architectural and implementation decisions.

Constraint Area	Description
Hardware	System must operate efficiently on low- to mid-range devices
Data Availability	Limited access to large, labeled mental health datasets
Performance	Near real-time processing under constrained server resources
AI Integration	Use of pre-trained, non-clinical sentiment analysis models
Scalability	Counselor workload balancing (e.g., one counselor managing multiple adolescents)

Table 7 Technical Constraints

2.6.4 Technology and Tool Constraints

The project is constrained to technologies that are appropriate for academic use and ethical deployment.

- Use of open-source or freely available software tools
- Supported programming languages and frameworks only
- Databases capable of handling both structured and unstructured data
- AI libraries limited to text analysis and trend detection

These constraints ensure cost-effectiveness, maintainability, and compliance with institutional policies.

2.6.5 Security Constraints

Security requirements impose additional design and implementation constraints, including:

- Mandatory encryption for all sensitive data at rest and in transit
- Secure authentication mechanisms for all users
- Strict role-based authorization enforcement
- Limited administrator access to user-generated content
- Protection against unauthorized access and data breaches

Security considerations are integrated across all system components.

2.6.6 Design Standards and Conventions

The NEURONET system adheres to established software engineering best practices, including:

- Modular and layered system architecture
- Clear separation of concerns between frontend, backend, and AI components
- Standard coding conventions and comprehensive documentation
- Secure API design principles
- Emphasis on maintainability, extensibility, and scalability

2.6.7 Data Retention and Deletion Constraints

Given the sensitive nature of adolescent mental health-related data and ethical compliance requirements, the following constraints govern data retention, archival, and deletion within the NEURONET system:

Retention Periods

- User account data (profiles, authentication records): Retained for the duration of active system use plus 1 year after account deactivation.
- Journal entries, mood records, and AI analysis results: Automatically anonymized after 2 years from creation date; full deletion after 5 years.
- Chat messages (counselor-adolescent, guardian-counselor): Retained for 3 years from last message date for reference and audit purposes.
- Consent records and audit logs: Retained for 7 years to comply with institutional and ethical review requirements.
- Alert and notification history: Retained for 2 years; aggregated thereafter for statistical use only.

Deletion Rules

- Upon guardian request, adolescent accounts may be deactivated immediately, rendering data inaccessible to all users except system administrators for compliance retention periods.
- Adolescents aged 18+ may request full account deletion, triggering automated data purging after a 30-day grace period.
- Backups containing sensitive data are encrypted and automatically purged after 90 days.
- Data scheduled for deletion is irreversibly anonymized or removed; no recovery is possible post-deletion.

Archival and Anonymization

- Data older than retention periods is automatically moved to archival storage and stripped of personally identifiable information (PII).
- Anonymized data may be used for aggregate trend analysis and system improvement but cannot be traced back to individual users.
- No raw journal content or private messages are retained in archival form.

Compliance Alignment

These constraints align with:

- Institutional ethical review board guidelines for adolescent data handling.
- General Data Protection Regulation (GDPR) principles for minors' data.
- Academic project data management policies.

2.7 Assumptions and Dependencies

This section outlines the key assumptions made during the development and operation of the NEURONET system and identifies dependencies on external technologies, infrastructure, and organizational factors. These assumptions and dependencies influence system design decisions, implementation strategies, and project planning. Any changes or violations of these assumptions may impact system functionality, performance, or project outcomes.

2.7.1 Assumptions

The development and operation of the NEURONET system are based on the following assumptions:

User Accessibility

It is assumed that adolescent users, guardians, counselors, and administrators have access to internet-enabled devices such as smartphones, laptops, or desktop computers capable of running the NEURONET application or web interface.

Internet Connectivity

The system assumes the availability of stable internet connectivity. Features such as journaling synchronization, AI-based sentiment analysis, dashboard updates, and role-based communication rely on continuous network access.

Guardian Participation and Consent

It is assumed that guardians will actively participate in the system by registering adolescent users, managing consent, and approving counselor communication when required. The system's operation depends on guardian engagement for ethical compliance.

Adolescent Engagement

The effectiveness of AI-based trend detection relies on regular adolescent interaction with the system. It is assumed that adolescents will actively use journaling, mood tracking, and reflection features over time.

AI Interpretation Limitations

AI and Natural Language Processing models are assumed to provide **indicative emotional trends**, not precise or error-free interpretations. It is assumed that users understand AI outputs as supportive insights rather than diagnostic or advisory conclusions.

Counselor Availability

It is assumed that a sufficient number of counselors will be available to support assigned adolescents and that each counselor can reasonably manage multiple adolescents (e.g., up to approximately 20) based on system workload balancing.

Requirement Stability

It is assumed that core system requirements will remain stable after formal approval, with only minor refinements occurring during development.

Ethical System Usage

It is assumed that all users will interact with the system ethically and in accordance with stated usage guidelines, privacy policies, and the system's intended non-clinical purpose.

2.7.2 Technical Dependencies

The NEURONET system depends on the availability and proper functioning of the following technical components and tools:

Dependency Area	Description
AI and NLP Libraries	Dependence on third-party, open-source Python libraries for sentiment and text analysis
Backend Frameworks	Use of Django or FastAPI for server-side logic and API services
Frontend Frameworks	Flutter for mobile interfaces and Next.js for web-based dashboards
Database Systems	PostgreSQL and MongoDB for structured and unstructured data storage
API Services	RESTful APIs for communication between system components

Table 8 Technical Dependencies

Any incompatibility or discontinuation of these tools may require system adaptation.

2.7.3 Infrastructure Dependencies

The system relies on external infrastructure services to ensure reliable operation:

- Cloud hosting services for deployment and scalability
- Secure servers for data storage and AI processing
- Automated backup and recovery mechanisms to prevent data loss
- Monitoring and logging tools to maintain system reliability and performance

Failure or unavailability of these infrastructure components may affect system accessibility, responsiveness, or data integrity.

2.7.4 Organizational and Academic Dependencies

As an academic capstone project, NEURONET is dependent on several organizational and institutional factors:

- Continuous academic supervision and feedback
- Access to university-provided development resources and tools
- Approval of documentation, milestones, and deliverables by the department
- Compliance with academic schedules, submission deadlines, and evaluation criteria

Changes in academic requirements or timelines may influence development progress.

2.7.5 External and Environmental Dependencies

External factors beyond the development team's control may impact the system:

- Updates or changes in third-party libraries, frameworks, or APIs
- Operating system or platform updates affecting compatibility
- Availability and quality of technical documentation and learning resources
- Changes in legal, ethical, or institutional guidelines related to data privacy and AI usage

Such changes may necessitate system maintenance or redesign.

2.7.6 Risk Considerations

If any assumptions are invalidated or dependencies fail, potential risks include:

- Reduced accuracy or reliability of AI-generated insights
- Decreased user engagement or usability
- Delays in development, testing, or deployment
- Security vulnerabilities or compliance issues
- Requirement for scope adjustment or architectural changes

To mitigate these risks, the NEURONET system adopts a modular architecture, clear role separation, continuous review, and adherence to established software engineering best practices.

CHAPTER THREE: SYSTEM REQUIREMENTS

3.1 Functional Requirements

Functional requirements describe the specific features and functions that the NEURONET system must provide to enable users to achieve their intended goals. These requirements define what the system shall do and outline the expected system behavior from the user's perspective. All functional requirements in this section are aligned with the system objectives, scope, user classes, and business rules.

Each functional requirement is written in a clear, simple, and unambiguous manner using the keyword "shall" to indicate mandatory system behavior. The requirements are grouped by feature areas to improve readability, traceability, and organization.

3.1.1 User Registration and Authentication

This feature ensures secure, role-based access to the system and enforces guardian-controlled registration for adolescent users.

FR-01: The system shall require guardians to initiate and complete adolescent user account registration on behalf of minors.

FR-02: The system shall require guardians to provide valid identification and contact information during adolescent registration.

FR-03: The system shall allow adolescent users to activate their account by setting a personal password using guardian-provided credentials.

FR-04: The system shall allow counselors to register accounts subject to administrative approval and verification of qualifications, specialization, professional affiliation, and a valid institutional email address.

FR-05: The system shall require all users to authenticate using a valid username and password before accessing system functionality.

FR-06: The system shall enforce role-based authentication to differentiate access privileges among user classes.

FR-07: The system shall deny access when authentication fails.

FR-08: The system shall allow authenticated users to securely log out of the system.

3.1.2 User Profile Management

This feature allows users to manage permitted personal information.

FR-09: The system shall allow users to view their profile information.

FR-10: The system shall allow users to update editable profile details such as contact information and preferences.

FR-11: The system shall restrict modification of sensitive identity information such as role assignments and guardian relationships.

FR-12: The system shall securely store all user profile data in the database.

FR-13: The system shall allow counselors to create and manage public educational pages (articles, tips, coping strategies).

FR-14: The system shall allow adolescents to follow counselor pages.

FR-15: The system shall allow guardians to view which counselor pages their adolescent is following.

Operational Description of Counselor Pages

Page Discovery and Visibility

Counselor-created educational pages are discoverable through a browsable directory within the system. Adolescents can search, filter, or browse pages by topic, counselor name, or popularity. Pages are publicly listed but require explicit follow action by the adolescent to appear in their personal feed.

Follow/Unfollow Behavior

Adolescents may choose to follow or unfollow counselor pages at any time without requiring guardian consent. Following a page enables the page's content to appear in the adolescent's personalized feed. Unfollowing removes the page from the feed but does not delete historical interaction records. The system tracks follow status for reporting and guardian visibility purposes.

Guardian Visibility

Guardians may view a list of counselor pages their adolescent is following through their guardian dashboard. This list includes page titles, counselor names, and follow dates, but does not include access to page content or adolescent interactions (reactions/comments) unless publicly visible within channel rules.

3.1.3 Consent Management

Consent management ensures ethical data handling and controlled access to adolescent information.

FR-16: The system shall require guardian consent before enabling analysis or sharing of adolescent data.

FR-17: The system shall allow guardians to grant or revoke consent for counselor access to summarized data only.

FR-18: The system shall enforce consent validation before allowing guardians or counselors to access any adolescent-related insights.

FR-19: The system shall allow adolescents to view their current consent status.

FR-20: The system shall record consent history for accountability and auditing purposes.

3.1.4 Journaling and Mood Reflection

This feature enables adolescents to privately express emotions and experiences.

FR-21: The system shall allow adolescent users to create personal journal entries.

FR-22: The system shall allow adolescents to record mood reflections at any time.

FR-23: The system shall allow adolescents to view their own journal history and mood trends.

FR-24: The system shall securely store journal and mood data and associate it with the correct user account.

FR-25: The system shall prevent guardians and counselors from accessing raw journal content.

Journal History and Mood Viewing Behavior

Adolescent users may access their journal history and mood reflections through a dedicated viewing interface after authentication. Journal entries and mood records are presented in chronological order, allowing adolescents to review past entries and observe changes in emotional patterns over time. Historical journal entries and mood data are displayed in a read-only format and cannot be edited or deleted after submission.

Mood trends are derived from recorded mood reflections and summarized visually to support self-awareness. Trend views present aggregated information over predefined time periods and do not expose raw analytical scores or internal system calculations. Access to journal history and mood trends is restricted to the adolescent user only; guardians and counselors are prevented from viewing raw journal content in accordance with privacy and consent constraints.

3.1.5 AI-Based Informational Interaction and Analysis

This feature defines the role of AI as a non-clinical, informational component.

FR-26: The system shall provide an AI chat interface for adolescents to interact with the system.

FR-27: The AI shall respond to general system-related and informational queries.

FR-28: The AI shall provide emotionally neutral and supportive responses without judgment.

FR-29: The AI shall not provide medical advice, therapy, diagnosis, or treatment recommendations.

FR-30: The AI shall redirect advice-seeking or clinical questions to human counselors.

FR-31: The AI shall recommend relevant counselor informational pages when medium-risk emotional patterns are detected from sentiment scores and mood trends, and shall be limited to informational content without generating clinical advice, diagnoses, or automatic escalation.

FR-32: The system shall analyze journal entries and AI interactions to generate sentiment scores and emotional trends.

FR-33: The system shall present AI analysis results as non-diagnostic emotional indicators derived from sentiment scores and mood trends, without exposing raw journal content or clinical interpretation.

AI Trend Analysis Workflow

The system performs periodic analysis of adolescent journal entries, recorded mood reflections, and AI interaction context to generate sentiment indicators and emotional trends. Analysis is conducted automatically by the system and does not require user initiation. Only text and mood inputs explicitly provided within the system are processed; no external data sources are used.

Generated sentiment scores and emotional trends are aggregated over predefined time intervals to identify general emotional patterns. These results are presented in summarized visual or textual

form through dashboards and trend views. Raw analytical outputs, internal scoring values, and model-specific metrics are not exposed to users.

AI-generated insights are strictly non-diagnostic and informational. The system does not label emotional states, predict conditions, or infer clinical outcomes. When medium-risk emotional patterns are identified, the system may recommend relevant counselor pages as supportive resources, while ensuring that interpretation and follow-up remain the responsibility of human guardians or counselors in accordance with consent settings.

AI Processing and Analysis Details

Data Processed by AI

The AI system processes only textual content entered by adolescent users within the following sources:

- Journal entry text
- Mood reflection notes
- AI chat session transcripts (user queries only)
- No external data sources, multimedia content, or metadata (location, time) are used for sentiment or trend analysis.

Constraints on AI Inputs

- Input text must be in English.
- Minimum journal entry length for analysis: 10 characters.
- Maximum input length per analysis cycle: 5,000 characters.
- AI does not process emojis, images, audio, or video content.
- Content containing personally identifiable information (PII) is anonymized before processing.

NLP Model Limitations

- Sentiment analysis is performed using pre-trained, open-source NLP models (e.g., VADER, TextBlob, or transformer-based models fine-tuned on non-clinical datasets).
- Models are trained on general English corpora, not on clinical or diagnostic mental health data.
- Accuracy is estimated at 75–85% for sentiment polarity detection in adolescent conversational text.
- Models do not support multilingual analysis, sarcasm detection, or contextual emotional nuance beyond basic polarity (positive/neutral/negative).

Frequency of Analysis

- Journal entries are analyzed once upon submission.
- Batch trend analysis runs nightly to aggregate emotional patterns over 7-, 14-, and 30-day windows.

- Real-time analysis is not performed; sentiment scores are available within 5 minutes of submission.

Handling of Accuracy Issues and Misinterpretations

- AI-generated sentiment scores are presented as “indicative trends” with a confidence disclaimer.
- Users are reminded that AI outputs are not clinical assessments.
- Misinterpreted or flagged content is logged for periodic manual review by system administrators.
- No automated alerts are generated solely based on low-confidence AI scores.

3.1.6 Counselor Communication and Assignment

This feature enables controlled professional support.

FR-34: The system shall allow adolescents to communicate privately with counselors only after guardian approval.

FR-35: The system shall assign counselors to adolescents based on system-defined allocation rules.

FR-36: The system shall limit the number of adolescents assigned to a counselor, with the maximum limit configurable by a system administrator.

FR-37: The system shall allow counselors to communicate with guardians when necessary without exposing private adolescent content.

FR-38: The system shall store counselor–adolescent communications securely.

3.1.7 Counselor Channels and Group Interaction

This feature supports guided, non-private group engagement.

FR-39: The system shall allow counselors to create informational channels.

FR-40: The system shall allow adolescents to join approved counselor channels.

FR-41: The system shall allow adolescents to react to and comment on counselor posts.

FR-42: The system shall restrict sharing of sensitive or personal disclosures within channels.

Operational Description of Counselor Channels

Channel Interaction Flow

Adolescents may discover channels through counselor profiles or curated channel listings. Joining a channel requires no guardian approval and grants read access to channel posts. Once joined, adolescents may view posts, react using predefined emoticons, and comment in compliance with channel guidelines. All interactions are logged and moderated; personal or clinically sensitive disclosures are automatically filtered and restricted.

Moderation and Restrictions

Counselors moderate their channels and may remove inappropriate comments or restrict users. The

system enforces content filters to prevent sharing of private journal excerpts, contact information, or explicit emotional disclosures. Channel interactions are visible to channel members and the managing counselor but are not included in guardian dashboards or alert systems.

3.1.8 Dashboards and Visualization

Dashboards present insights in a role-appropriate manner.

FR-43: The system shall provide adolescents with a personal dashboard displaying emotional trends.

FR-44: The system shall provide guardians with aggregated dashboards and alerts, subject to consent.

FR-45: The system shall provide counselors with dashboards for assigned adolescents only, showing consent summaries, emotional trend summaries, mood/activity charts (7/14/30 days), and session schedules.

FR-46: Counselors shall not have access to raw journal content, private AI chats, or guardian-only data through the dashboard.

FR-47: The system shall present data using clear visual elements such as charts and summaries.

Operational Description of Dashboards

Adolescent Dashboard Operation

The adolescent dashboard provides a personalized, read-only view of the user's emotional trends and mood activity. Adolescents can access the dashboard upon authentication and view summarized emotional indicators generated from their journal entries and mood reflections. The dashboard presents trends over predefined time ranges and does not allow modification of historical data. No alerts or counselor-specific information are displayed to adolescents through this interface.

Guardian Dashboard Operation

The guardian dashboard presents aggregated emotional insights and alert summaries related to the adolescent, strictly subject to active consent settings. Guardians can view high-level trend summaries and alerts without access to raw journal entries, private AI interactions, or detailed emotional content. Dashboard data is presented in an informational format only and does not support direct interaction with adolescent-generated content. Any changes to consent settings immediately affect the visibility of dashboard information.

Counselor Dashboard Operation

The counselor dashboard functions as a monitoring interface for assigned adolescents only. Counselors can view consent summaries, emotional trend overviews, mood and activity charts for predefined periods (7, 14, and 30 days), and scheduled session information. The dashboard allows filtering by time range but restricts access to raw journal content, private AI conversations, and

guardian-only data. All information presented is non-diagnostic and intended solely to support professional awareness and follow-up.

3.1.9 Alerts and Notifications

This feature supports early awareness and follow-up.

FR-48: The system shall generate alerts based on predefined alert thresholds, where threshold values are configurable by a system administrator.

FR-49: The system shall notify authorized guardians and counselors based on consent settings.

FR-50: The system shall allow authorized users to view alert history.

FR-51: The system shall avoid generating alerts that imply diagnosis or emergency decisions.

Alert Generation and Review Workflow

When predefined emotional thresholds are reached, the system generates informational alerts and delivers notifications to authorized guardians and counselors according to active consent settings. Alerts are intended to promote awareness and follow-up rather than immediate or clinical intervention.

Authorized users can access an alert review interface to view current and historical alerts in chronological order. Each alert includes a timestamp, associated adolescent identifier (where permitted), and a brief non-diagnostic description of the triggering condition. Users may review alerts for informational purposes only; alerts cannot be modified, escalated automatically, or interpreted as medical or emergency determinations.

Alert history is retained to support monitoring over time and accountability. The system does not generate language suggesting diagnosis, urgency, or required action. Any follow-up actions are performed externally by guardians or counselors in accordance with professional judgment and consent constraints.

3.1.10 Administrative Management

Administrative functions support system operation.

FR-52: The system shall allow administrators to manage user accounts and roles.

FR-53: The system shall allow administrators to assign counselors to adolescents.

FR-54: The system shall allow administrators to monitor system performance and logs.

FR-55: The system shall restrict administrators from accessing private user content.

3.1.11 Data Security and Access Control

These requirements enforce system security.

- FR-56:** The system shall enforce role-based access control for all system features.
- FR-57:** The system shall ensure that sensitive data is accessible only to authorized users.
- FR-58:** The system shall log significant system actions for monitoring and auditing purposes.

3.2 Use Case Model

The Use Case Model describes how different user classes interact with the NEURONET system to achieve specific goals. It provides a functional view of the system by identifying system actors and the use cases associated with each actor. The model clarifies system boundaries, user responsibilities, and access control while ensuring that all functional requirements are represented through user interactions.

3.2.1 Use Case Diagram

The NEURONET system includes the following primary actors:

- Adolescent User
- Guardian
- Counselor
- System Administrator

Each actor interacts with the system according to their role and assigned permissions.

Adolescent User Use Cases

- Activate Account
- Log In / Log Out
- Manage Profile
- View Consent Status
- Write Journal Entry
- Record Mood Reflection
- Interact with AI (Informational)
- View Personal Dashboard
- View Alerts
- Chat with Counselor (With Approval)
- Join Counselor Channels
- React and Comment on Channel Posts

Figure 2.5: Adolescent User Use Case Diagram

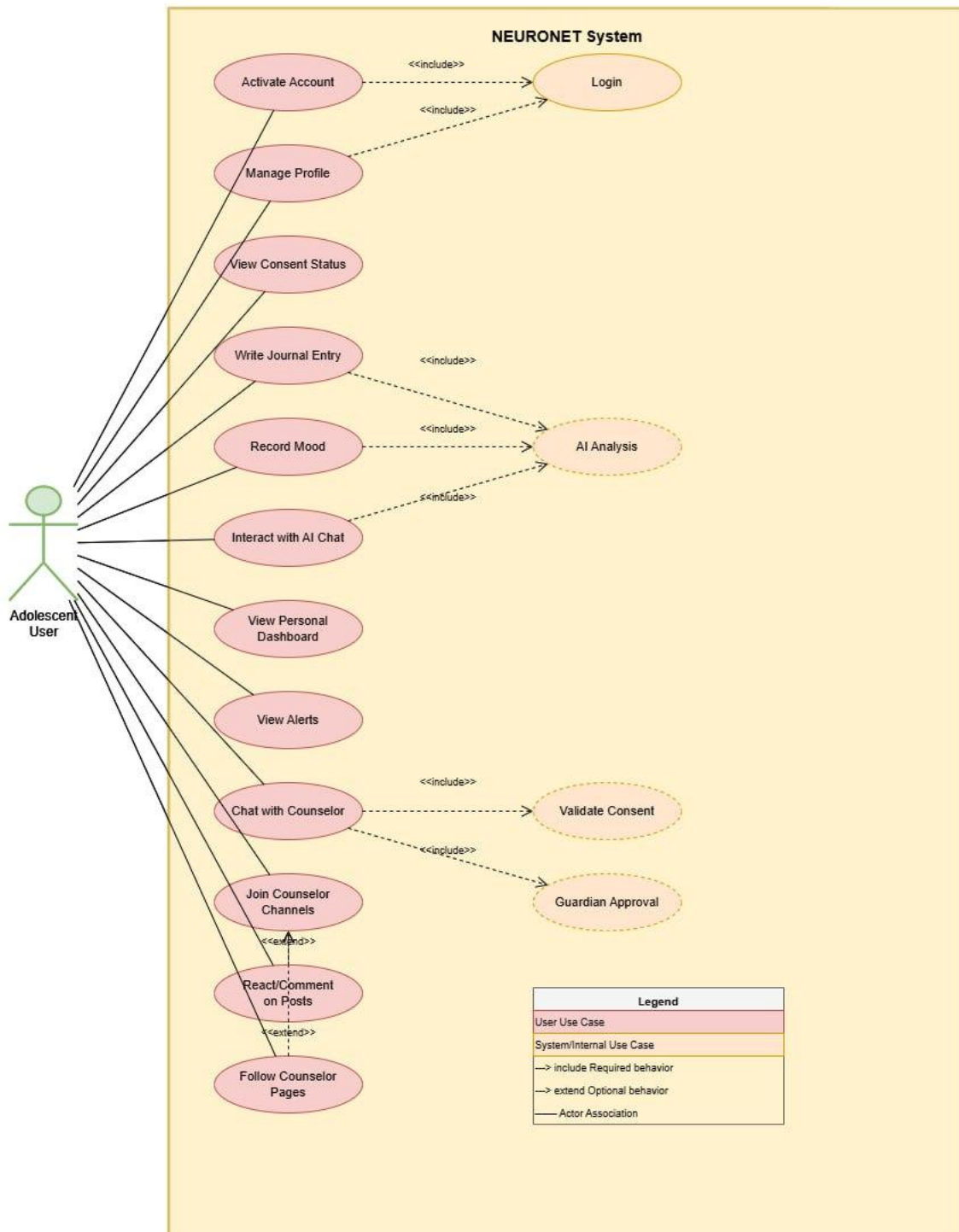


Figure 1 Adolescent User Use Cases Diagram

Guardian Use Cases

- Register Adolescent
- Log In / Log Out
- Manage Profile
- Grant or Revoke Consent
- View Aggregated Dashboard
- Receive Alerts
- Communicate with Counselor

Figure 2.6: Guardian Use Case Diagram

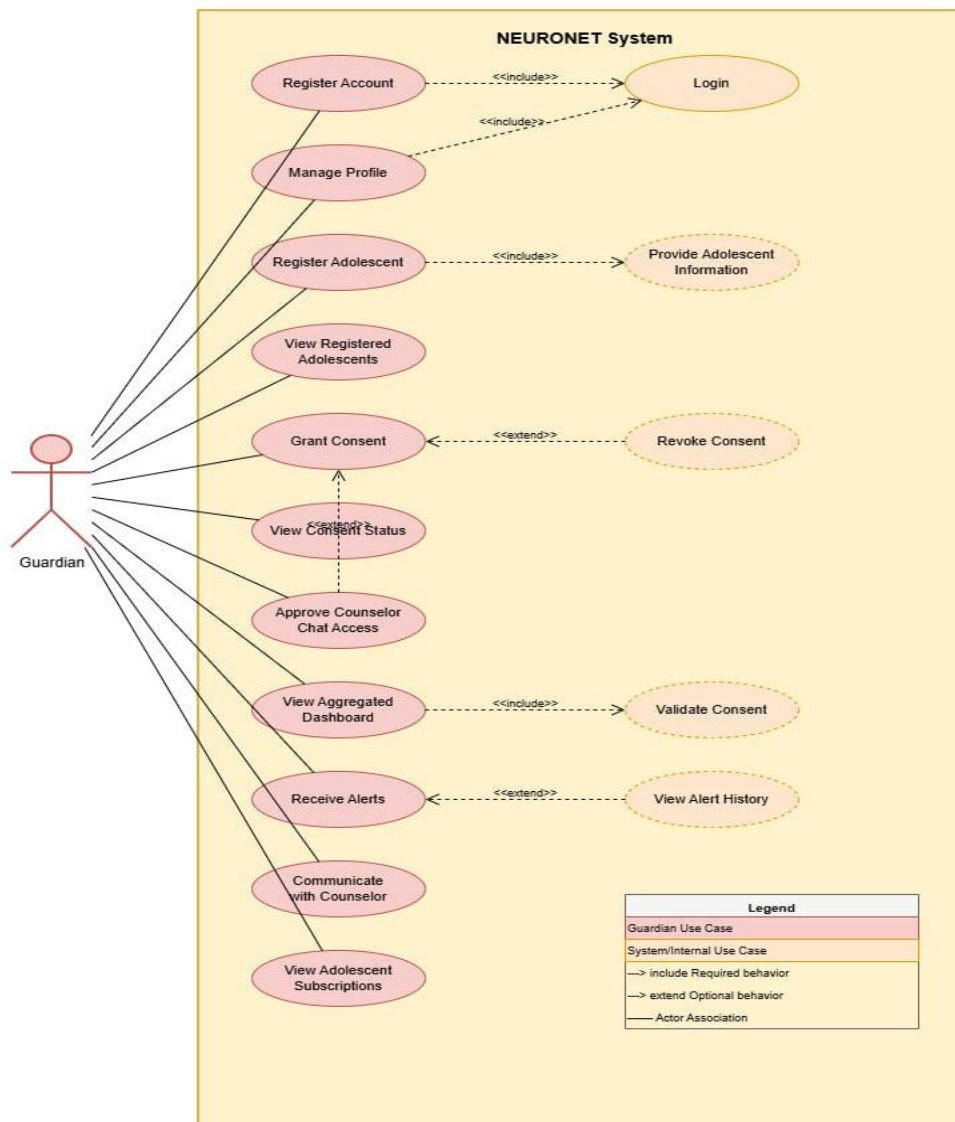


Figure 2 Guardian Use Cases Diagram

Counselor Use Cases

- Log In / Log Out
- Manage Profile
- View Assigned Adolescents
- View Behavioral Trends
- Receive Alerts
- Chat with Adolescent
- Communicate with Guardian
- Create and Manage Channels

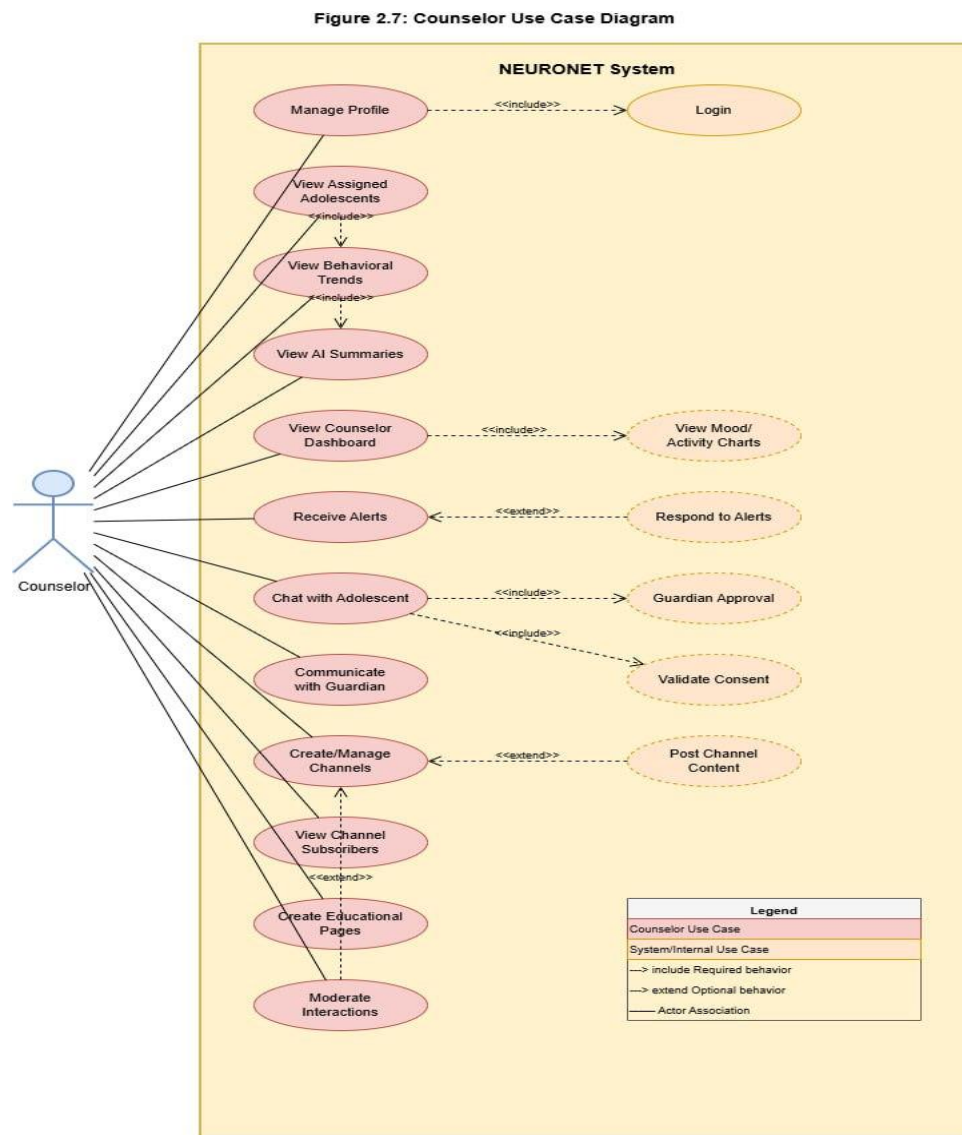


Figure 3 Counselor Use Cases Diagram

Figure 4 System Administrator Use Cases Diagram

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```

    usecaseDiagram
        actor SA as System Administrator
        usecase MUA as Manage User Accounts
        usecase MRP as Manage Roles and Permissions
        usecase ACA as Assign Counselors to Adolescents
        usecase MSA as Monitor System Activity
        usecase MSC as Manage System Configuration
        usecase CO as Consent Oversight
        usecase GR as Generate Reports
        usecase MBR as Manage Backup and Recovery
        usecase L as Login
        usecase CUA as Create User Account
        usecase DSA as Deactivate/Suspend Account
        usecase AR as Assign Role
        usecase WB as Workload Balancing
        usecase VAL as View Audit Logs
        usecase CAP as Configure AI Parameters
        usecase SAT as Set Alert Thresholds

        SA --- MUA
        SA --- MRP
        SA --- ACA
        SA --- MSA
        SA --- MSC
        SA --- CO
        SA --- GR
        SA --- MBR

        MUA -.-> L : <<include>>
        MUA -.-> CUA : <<include>>
        MUA -.-> DSA : <<include>>
        MUA -.-> AR : <<include>>
        MUA -.-> WB : <<include>>
        MUA -.-> VAL : <<include>>
        MUA -.-> CAP : <<include>>
        MUA -.-> SAT : <<include>>

        MRP -.-> CUA : <<extend>>
        MRP -.-> AR : <<include>>
        MRP -.-> WB : <<include>>
        MRP -.-> VAL : <<include>>
        MRP -.-> CAP : <<include>>
        MRP -.-> SAT : <<include>>

        MSC -.-> MSA : <<include>>
        MSC -.-> WB : <<include>>
        MSC -.-> VAL : <<include>>
        MSC -.-> CAP : <<include>>
        MSC -.-> SAT : <<include>>

        MSC -.-> MSC : <<extend>>
    
```

NEURONET System

Legend

- Administrator Use Case
- System/Internal Use Case
- .-> include Required behavior
- .-> extend Optional behavior
- Actor Association

Figure 2.4: NEURONET Complete System Use Case Diagram

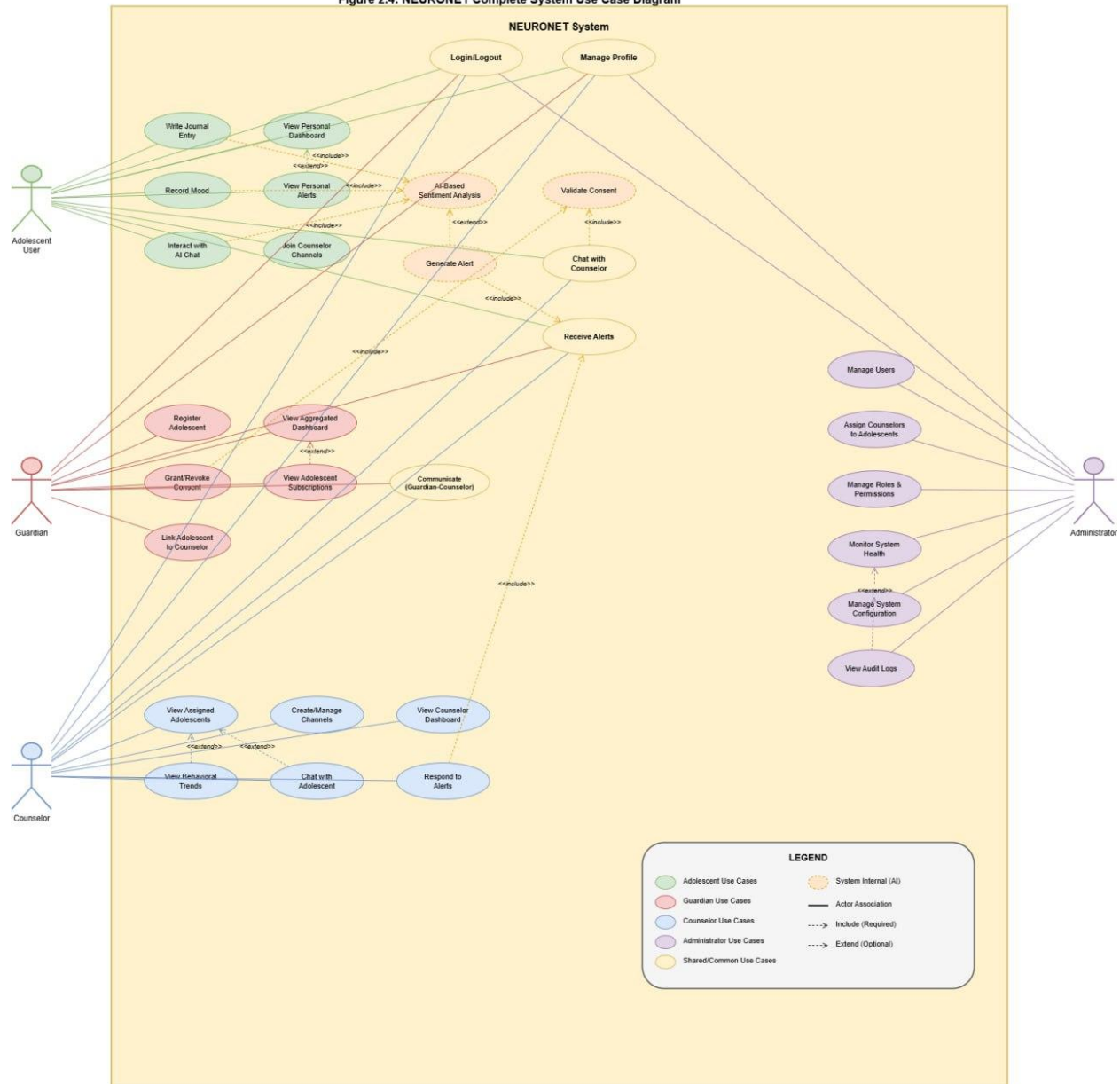


Figure 5 illustrates the Use Case Diagram of the NEURONET system, showing interactions between system actors and major system functionalities.

Use Case Descriptions

Use Case 1: Register Adolescent

Item	Description
Use Case ID	UC-01
Actor	Guardian
Description	Registers an adolescent account on behalf of a minor
Precondition	Guardian is authenticated
Postcondition	Adolescent account is created
Main Flow	1. Guardian opens registration page 2. Guardian enters adolescent details 3. System validates information 4. Account is created
Alternative Flow	Validation failure results in error message

Table 9 Use case 1-Register Adolescent

Use Case 2: User Authentication

Item	Description
Use Case ID	UC-02
Actor	All Users
Description	Authenticates users into the system
Precondition	User account exists
Postcondition	User gains authorized access
Main Flow	1. User enters credentials 2. System verifies credentials 3. Access granted
Exception Flow	Invalid credentials deny access

Table 10 User Case 2-User Authentication

Use Case 3: Write Journal Entry

Item	Description
Use Case ID	UC-03
Actor	Adolescent User
Description	Records personal reflections
Precondition	User authenticated
Postcondition	Journal entry saved
Main Flow	1. User writes entry 2. System saves securely
Alternative Flow	User cancels entry

Table 11 Use Case 3-Write Journal Entry

Use Case 4: Interact with AI

Item	Description
Use Case ID	UC-04
Actor	Adolescent User
Description	Interacts with AI for system-related and informational queries
Precondition	User authenticated
Postcondition	AI response displayed
Main Flow	1. User submits query 2. AI responds informationally
Constraint	AI does not provide advice or diagnosis

Table 12 Use Case 4-Interact with AI

Use Case 5: Chat with Counselor

Item	Description
Use Case ID	UC-05
Actor	Adolescent User, Counselor
Description	Enables private communication
Precondition	Guardian consent granted
Postcondition	Messages exchanged
Main Flow	1. User opens chat 2. Messages sent and received

Table 13 Use Case 5-Chat with Counselor

Use Case 6: View Dashboard

Item	Description
Use Case ID	UC-06
Actor	Adolescent, Guardian, Counselor
Description	Displays role-based summaries with filtering and trend visualization
Precondition	User authenticated and authorized for dashboard access

Item	Description
Main Flow	1. User selects dashboard from navigation 2. System validates role and consent settings 3. System retrieves relevant data (trends, alerts, summaries) based on user role 4. User may apply filters (e.g., time range: 7/14/30 days) 5. System displays visualizations and summaries
Postcondition	Dashboard displayed with applicable data filters applied

Table 14 Use Case 6-View Dashboard

Use Case 7: Receive Alert

Item	Description
Use Case ID	UC-07
Actor	Guardian, Counselor
Description	Notifies concerning trends based on configurable thresholds and consent rules
Precondition	Consent granted, AI analysis completed, emotional threshold met
Postcondition	Alert delivered to authorized roles, logged in system
Main Flow	1. AI analysis detects emotional pattern meeting predefined threshold 2. System validates consent settings for data sharing 3. Alert is generated with severity level and trend summary 4. Notification is sent to guardian and/or counselor based on role permissions 5. Alert appears in recipient's dashboard and alert history

Table 15 Use Case 7-Receive Alert

Use Case 8: Administrative Management

Item	Description
Use Case ID	UC-08
Actor	System Administrator
Description	Manages system configuration
Precondition	Admin authenticated
Postcondition	System updated
Main Flow	1. Admin selects action 2. System applies changes

Table 16 Use Case 8-Administrative Management

Use Case 9: Join & Interact with Counselor Channel

Item	Description
Use Case ID	UC-09
Actor	Adolescent
Description	Join counselor-created channels and interact with posts via reactions and comments
Precondition	Adolescent authenticated, channel is active and joinable
Postcondition	Interaction recorded; comment/reaction visible in channel
Main Flow	1. Adolescent browses available channels 2. Selects a channel and chooses to join 3. Views counselor posts within the channel 4. May react using predefined emoticons or add a comment 5. System validates comment against sensitivity filters 6. Comment/reaction is published if approved

Table 17 Use Case 9: Join & Interact with Counselor Channel

3.2.2 Summary of Use Case Model

The use case model defines system behavior through actor interactions while enforcing privacy, consent, and non-clinical constraints. It ensures functional completeness and traceability across system requirements.

3.3 Data Requirements

This section describes the data that the NEURONET system must handle, store, process, and generate in order to perform its required functions. Proper definition of data requirements ensures data consistency, integrity, security, and traceability throughout the system lifecycle. Due to the sensitive nature of adolescent mental well-being data, strict emphasis is placed on privacy, consent, and ethical handling.

3.3.1 Overview

The NEURONET system manages both user-generated and system-generated data.

User-generated data includes journal entries, mood records, chat messages, channel interactions, and consent preferences.

System-generated data includes AI sentiment scores, emotional trend summaries, alerts, counselor assignments, and system logs.

All data shall be securely stored and accessed according to role-based access control and consent rules. The system shall support structured and semi-structured data formats and ensure that no raw sensitive content is exposed to unauthorized users.

3.3.2 Entities, Attributes, and Relationships

1. User Entity

Description: Base entity storing common information for all system users.

- Attributes:
- UserID (Primary Key)
- FullName
- Email
- PasswordHash
- Role (Adolescent, Guardian, Counselor, Administrator)
- CreatedAt
- AccountStatus (Active/Inactive/Suspended)

Relationships:

- One User can send many Chat Messages (as Sender)
- One User can receive many Chat Messages (as Receiver)

2. Guardian Entity

Description: Extends User for guardian role.

Attributes:

- GuardianID (Primary Key, Foreign Key to User.UserID)

Relationships:

- One Guardian can have many GuardianAdolescentLinks
- One Guardian can grant many Consents

3. Adolescent Entity

Description: Extends User for adolescent role.

Attributes:

- AdolescentID (Primary Key, Foreign Key to User.UserID)
- DateOfBirth (nullable)

Relationships:

- One Adolescent can have many Journal Entries
- One Adolescent can have many Mood Records
- One Adolescent can have many AI Analysis Records
- One Adolescent can have many Alerts
- One Adolescent can have many GuardianAdolescentLinks
- One Adolescent can have many ChannelMemberships
- One Adolescent can have many ChannelInteractions
- One Adolescent can have many Consent Records
- One Adolescent can be assigned to one Counselor via CounselorAssignment

4. Counselor Entity

Description: Extends User for counselor role.

Attributes:

- CounselorID (Primary Key, Foreign Key to User.UserID)
- ProfessionalAffiliation (nullable)
- InstitutionEmail (nullable)

- VerificationStatus (Pending/Approved/Rejected, default Pending)

Relationships:

- One Counselor can create many Channels
- One Counselor can create many ChannelPosts
- One Counselor can be assigned to many Adolescents via CounselorAssignment
- One Counselor can act on many Alerts
- One Counselor can moderate many ChannelInteractions

5. Admin Entity

Description: Extends User for administrator role.

Attributes:

- AdminID (Primary Key, Foreign Key to User.UserID)

Relationships:

- One Admin can assign many CounselorAssignments (as AssignedBy)

6. GuardianAdolescentLink Entity

Description: Explicitly links guardians to adolescents they registered.

Attributes:

- LinkID (Primary Key)
- GuardianID (Foreign Key to Guardian.GuardianID)
- AdolescentID (Foreign Key to Adolescent.AdolescentID)
- RelationshipType (Parent/LegalGuardian/Other)
- IsPrimary (boolean, default false)
- LinkedAt
- IsActive

Relationships:

- One Guardian can have multiple Adolescent links
- One Adolescent can have multiple Guardian links

Constraint: Only ONE primary guardian per adolescent (IsPrimary = true)

7. Journal Entry Entity

Description: Stores private reflections written by adolescent users.

Attributes:

- JournalID (Primary Key)
- AdolescentID (Foreign Key to Adolescent.AdolescentID)
- Content
- CreatedAt
- SentimentScore (System Generated, nullable)
- LastAnalyzedAt (timestamp of last AI analysis)

Relationships:

- Many Journal Entries belong to one Adolescent

8. Mood Record Entity

Description: Stores mood check-ins recorded by adolescents.

Attributes:

- MoodID (Primary Key)
- AdolescentID (Foreign Key to Adolescent.AdolescentID)
- MoodType (from predefined enum: Happy/Neutral/Sad/Anxious/Excited/etc.)
- Intensity (1-5 scale, nullable)
- RecordedAt
- ContextNotes (optional brief context, nullable)

Relationships:

- Many Mood Records belong to one Adolescent

9. Chat Message Entity

Description: Stores private text-based communication messages.

Attributes:

- MessageID (Primary Key)
- SenderID (Foreign Key to User.UserID)
- ReceiverID (Foreign Key to User.UserID)
- MessageContent
- Timestamp
- MessageType (CounselorChat / GuardianChat / AdminAlert)
- ConversationID (groups messages in a conversation)
- ReadStatus (Read/Unread)

Relationships:

- Messages are associated with Users as sender and receiver

Constraints:

- CounselorChat only between Counselor and Adolescent
- GuardianChat only between Guardian and Counselor

10. Consent Entity

Description: Records guardian-controlled consent for data sharing and communication.

Attributes:

- ConsentID (Primary Key)
- AdolescentID (Foreign Key to Adolescent.AdolescentID)
- GuardianID (Foreign Key to Guardian.GuardianID - MUST be primary guardian)
- ConsentType (JournalAnalysis / CounselorCommunication / DataSharing / AlertNotification / ChannelParticipation)
- GrantedToRole (Guardian / Counselor / Both)
- ConsentStatus (Granted/Denied/Revoked)
- GrantedAt
- RevokedAt
- ExpiryDate (nullable)
- Notes (optional reason for revocation)

Relationships:

- One Adolescent can have multiple Consent Records

Constraint: Only primary guardians can grant consent for adolescents

11. AI Analysis Entity

Description: Stores AI-generated sentiment indicators and emotional trends.

Attributes:

- AnalysisID (Primary Key)
- AdolescentID (Foreign Key to Adolescent.AdolescentID)
- AnalysisType (Sentiment / Trend / Pattern)
- ResultSummary
- ConfidenceScore (0.0-1.0)
- GeneratedAt
- DataRangeStart (what time period was analyzed)

- DataRangeEnd

Relationships:

- One AI Analysis Record belongs to one Adolescent

12. Alert Entity

Description: Stores system-generated alerts based on AI-detected thresholds.

Attributes:

- AlertID (Primary Key)
- AdolescentID (Foreign Key to Adolescent.AdolescentID)
- AlertType (EmotionalPattern / MoodDrop / JournalFrequency / ContentFlag)
- SeverityLevel (Low/Medium/High)
- TriggerDescription (detailed description of what triggered)
- TriggeringData (JSON of data that triggered - optional)
- CreatedAt
- ViewedStatus (boolean)
- ActionStatus (Pending/Reviewed/InProgress/Resolved/Escalated)
- ActionTakenBy (Foreign Key to Counselor.CounselorID - counselor who acted, nullable)
- ActionDate (when action was taken, nullable)
- ActionNotes (what was done, nullable)

Relationships:

- One Adolescent can have many Alerts
- Alert can be acted upon by one Counselor

13. Counselor Assignment Entity

Description: Manages assignment of counselors to adolescents.

Attributes:

- AssignmentID (Primary Key)
- CounselorID (Foreign Key to Counselor.CounselorID)
- AdolescentID (Foreign Key to Adolescent.AdolescentID)
- AssignedAt
- AssignmentStatus (Active/Inactive/Pending)
- AssignedBy (Foreign Key to Admin.AdminID - admin who made assignment)
- Notes (optional assignment reason)

Relationships:

- One Counselor can be assigned to many Adolescents
- One Adolescent is assigned to one Counselor (at a time)
- Constraint: Counselor assignment limit (e.g., max 20 per counselor)

14. Channel Entity

Description: Unified informational and educational spaces created by counselors.

Attributes:

- ChannelID (Primary Key)
- CounselorID (Foreign Key to Counselor.CounselorID)
- ChannelName
- Description
- ChannelType (Educational/Supportive/Discussion/ResourceLibrary)
- ContentCategory (Articles/Tips/CopingStrategies/General)
- CreatedAt
- IsActive
- AccessLevel (Public/ApprovedOnly/FollowersOnly)
- AllowComments (boolean)
- AllowReactions (boolean)
- IsFeatured (boolean - for important educational channels)

Relationships:

- One Counselor can create many Channels
- One Channel can have many ChannelPosts
- One Channel can have many ChannelMemberships

15. ChannelMembership Entity

Description: Tracks which adolescents are members/followers of which channels.

Attributes:

- MembershipID (Primary Key)
- ChannelID (Foreign Key to Channel.ChannelID)
- AdolescentID (Foreign Key to Adolescent.AdolescentID)
- JoinedAt
- MembershipType (Follower/Member/Subscriber)
- NotificationPreference (All/Important/None)
- GuardianNotified (boolean - tracks if guardian was informed)

Relationships:

- Many Memberships belong to one Channel
- Many Memberships belong to one Adolescent

16. ChannelPost Entity

Description: Individual posts within a channel (can be articles, tips, discussions).

Attributes:

- PostID (Primary Key)
- ChannelID (Foreign Key to Channel.ChannelID)
- CounselorID (Foreign Key to Counselor.CounselorID - who posted)
- Title
- Content
- PostType (Article/Tip/Resource/Question/DiscussionStarter)
- IsEducational (boolean - marks as formal educational content)
- CreatedAt
- UpdatedAt
- IsPinned (boolean)
- AllowComments (boolean - can override channel setting)
- ViewCount (tracks popularity)

Relationships:

- Many Posts belong to one Channel
- One Post can have many ChannelInteractions

17. ChannelInteraction Entity

Description: Stores adolescent interactions with channel posts.

Attributes:

- InteractionID (Primary Key)
- PostID (Foreign Key to ChannelPost.PostID)
- AdolescentID (Foreign Key to Adolescent.AdolescentID)
- InteractionType (Reaction / Comment / Save / Share)
- Content (for comments, nullable)
- ReactionType (Like/Support/Helpful/Question/etc. - for reactions, nullable)
- IsSaved (boolean - for bookmarking educational content)
- CreatedAt
- IsVisible (moderation flag, default true)
- ModeratedBy (Foreign Key to Counselor.CounselorID - counselor who moderated, nullable)

Relationships:

- Many Interactions belong to one ChannelPost
- Many Interactions belong to one Adolescent

3.3.3 Entity-Relationship Diagram (ERD)

The Entity-Relationship Diagram (ERD) represents the logical structure of the NEURONET database, showing relationships among users, journal entries, mood records, chat messages, consent records, AI analysis results, alerts, counselor assignments, and channels.

Key relationships include:

- One-to-many relationship between User and Journal Entry
- One-to-many relationship between User and Mood Record
- One-to-many relationship between User and Consent
- One-to-many relationship between User and AI Analysis
- One-to-many relationship between User and Alert
- One-to-many relationship between Counselor and Adolescent (via Assignment)
- One-to-many relationship between Counselor and Channel

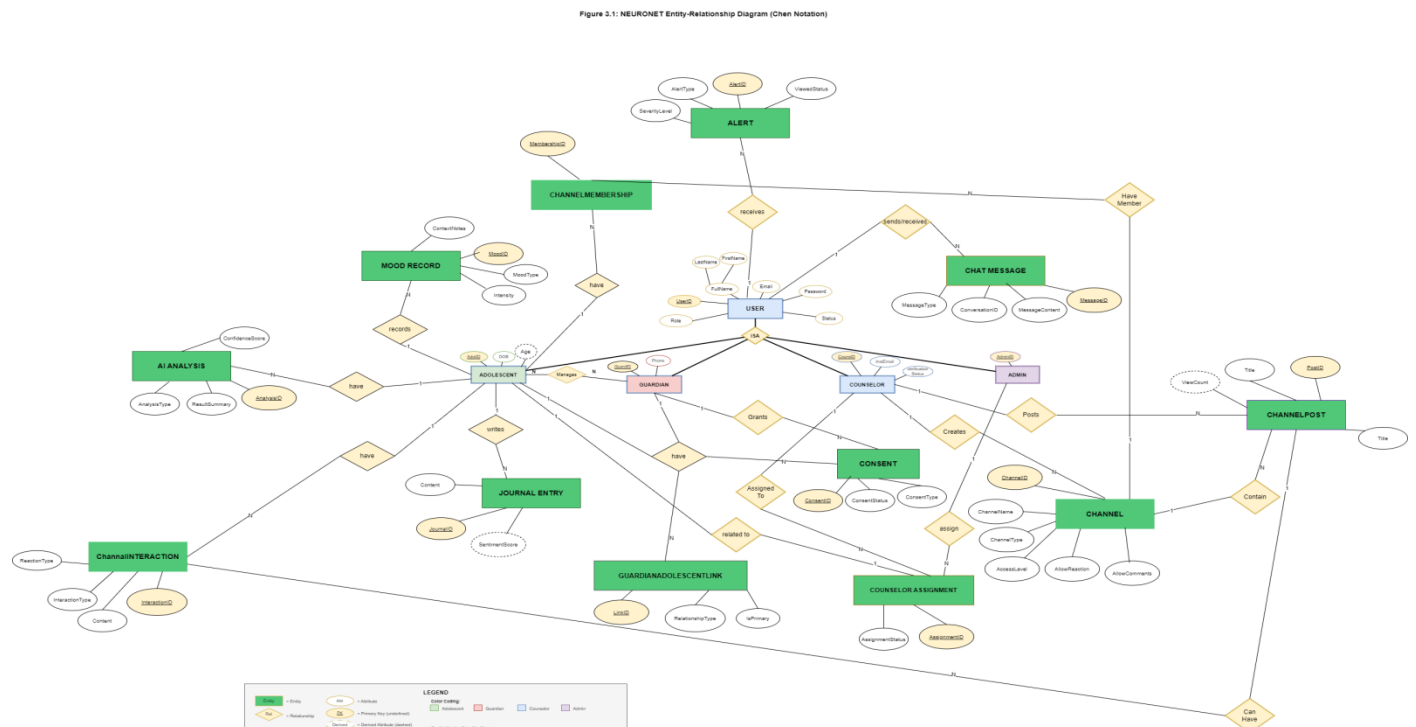


Figure 6 illustrates the Entity-Relationship Diagram (ERD) of the NEURONET system.

3.3.4 Data Validation Rules

Rule ID	Validation Rule Description
DR-01	Mandatory fields shall not be left empty
DR-02	Email addresses shall follow a valid format
DR-03	Passwords shall meet minimum security requirements
DR-04	Mood values shall be selected from predefined options
DR-05	Journal content length shall be within defined limits
DR-06	Consent status shall be explicitly recorded
DR-07	Duplicate user accounts shall not be allowed
DR-08	Counselor assignment limits shall be enforced
DR-09	AI analysis records shall not contain clinical diagnoses

Table 18 **Data Validation Rules**

3.4 External Interface Requirements

External interface requirements define how the NEURONET system interacts with users, software components, hardware, and communication networks. Although the system is not an embedded system, these interfaces are essential for interoperability, usability, security, and seamless system integration.

3.4.1 Hardware Interface

The NEURONET system interfaces with standard consumer hardware devices to ensure accessibility and ease of use.

- **Client Devices:**
 - Android smartphones for adolescent users
 - Desktop and laptop computers for guardians, counselors, and system administrators
- **Hardware Constraints:**
 - No specialized, external, or wearable hardware devices are required
 - The system operates on low- to mid-range devices without performance degradation

3.4.2 Software Interfaces

The system shall utilize a **RESTful API** architecture to facilitate secure communication between client applications (Frontend) and the server (Backend).

A. API Protocol and Data Formats

- **Communication Protocol:** All software interfaces shall communicate via **HTTPS** (TLS 1.3) using standard HTTP methods (GET, POST, PUT, PATCH, DELETE).
- **Data Interchange Format:** The primary format for data exchange shall be **JSON** (JavaScript Object Notation) encoded in UTF-8. All API responses must set the `Content-Type` header to `application/json`.
- **Authentication Interface:** All protected API endpoints shall require a valid **JSON Web Token (JWT)** passed in the HTTP `Authorization` header using the Bearer schema (`Authorization: Bearer <token>`).

B. Key API Endpoints and Specification

The system shall implement the following core API endpoints to support system functionality, as defined in the system design sequence diagrams:

Feature Area	HTTP Method	Endpoint Path	Input Payload (JSON)	Output Response (JSON)	Related Req
Auth	POST	/api/v1/auth/login	<pre>{ "email": "user@ex.com", "password": "..."} </pre>	<pre>{ "token": "jwt...", "role": "ADOLESCENT", "expires_in": 3600 } </pre>	FR-03, FR-05
Journaling	POST	/api/v1/journals	<pre>{ "title": "My Day", "content": "Today was...", "mood_id": "uuid" } </pre>	<pre>{ "journal_id": "uuid", "created_at": "timestamp", "status": "saved" } </pre>	FR-21
Analysis	GET	/api/v1/analyses/{id}	None	<pre>{ "sentiment_score": 0.8, "risk_level": "LOW", "generated_at": ... } </pre>	FR-32

				"timestamp" }	
--	--	--	--	---------------	--

Feature Area	HTTP Method	Endpoint Path	Input Payload (JSON)	Output Response (JSON)	Related Req
Chat	POST	/api/v1/chat/request	{ "counselor_id": "uuid", "adolescent_id": "uuid" }	{ "request_id": "uuid", "status": "pending_consent" }	FR-34
Consent	PUT	/api/v1/consent/{id}	{ "status": "GRANTED", "scope": ["alert_sharing"] }	{ "consent_id": "uuid", "updated_at": "timestamp" }	FR-17
Alerts	GET	/api/v1/alerts	?severity=HIGH	[{ "alert_id": "uuid", "message": "Trend detected...", "severity": "HIGH" }]	FR-50

Table 19 External Interface API Specifications

C. Database Interfaces

- **Relational Data Interface:** The backend services shall interface with **PostgreSQL** using an Object-Relational Mapper (ORM) for managing structured data entities, including Users, Consents, and Assignments.
- **Document Data Interface:** The backend shall interface with **MongoDB** using a native driver for managing unstructured and high-volume data, including Journal Entries, Chat Logs, and AI Analysis results.

D. Notification Service Interface

- **Push Notification Interface:** The system shall interface with an external **Push Notification Service (e.g., Firebase Cloud Messaging)** via HTTPS to deliver real-time alerts to mobile devices.

- **Email Service Interface:** The system shall interface with an SMTP server (TLS enabled) to send transactional emails for account verification and critical security alerts.

3.4.3 User Interface

User interface requirements ensure accessibility, emotional safety, and clarity for all user roles.

- The system shall provide a **mobile-friendly interface** optimized for adolescent users
- The system shall provide **web-based dashboards** for guardians, counselors, and administrators
- The interface shall display **role-specific content** based on user permissions
- The system shall use **simple, supportive, and non-technical language**
- Alerts and insights shall be presented in a **non-judgmental and non-alarming manner**
- Navigation shall be consistent across all platforms

The user interface design prioritizes usability, trust, and emotional comfort.

3.4.4 Communication Interface

The NEURONET system uses secure communication mechanisms to protect sensitive data.

- RESTful APIs for internal communication between system components
- HTTPS protocol for all client-server data transmission
- Encrypted data exchange to prevent unauthorized access
- Notification services for delivering alerts and system messages

These interfaces ensure secure, reliable, and efficient interaction across the system.

3.4.5 Error Handling Interface

To ensure consistent integration and debugging, the system shall utilize a standardized error response catalog for all external interfaces:

- **Standard Error Schema:** All API error responses shall adhere to the following JSON structure:

```
{
  "status": 401,
  "error_code": "UNAUTHORIZED",
  "message": "Invalid credentials provided.",
  "details": "Password does not match records.",
  "timestamp": "2025-12-19T10:00:00Z"
}
```

}

- **HTTP Status Mapping:** The interface shall return standard HTTP status codes to indicate operation outcomes:
 - 200 OK: Successful synchronous request.
 - 201 Created: Resource successfully created.
 - 400 Bad Request: Validation failure (e.g., missing fields).
 - 401 Unauthorized: Missing or invalid JWT token.
 - 403 Forbidden: User authenticated but lacks role permission (e.g., Guardian trying to read Journal).
 - 500 Internal Server Error: Unhandled system failure.

3.4.5.1 Error Message Catalog (Representative)

The backend shall return standardized error responses in JSON with fields { "error_code": "STRING", "message": "Human readable" }. Representative mapping:

HTTP Status	error_code	Message (human)	When returned
400	BAD_REQUEST	Invalid input; check fields	Payload schema fails validation
401	UNAUTHORIZED	Authentication required or invalid	Missing/invalid token
403	FORBIDDEN	Access denied	Role/consent violation
404	NOT_FOUND	Resource not found	Missing id
409	CONFLICT	Resource already exists or idempotency conflict	Duplicate client_entry_id differing content
413	PAYLOAD_TOO_LARGE	Submitted content exceeds limit	entry_text > 2000 chars
422	UNPROCESSABLE	Business rule violation	Consent or policy breach
429	TOO_MANY_REQUESTS	Rate limit exceeded	Request throttling
500	SERVER_ERROR	Internal server error	Transient/system failure
503	SERVICE_UNAVAILABLE	Service temporarily unavailable	Maintenance/outage

Table 20 API Error Code Catalog and Handling Mapping

Clients shall present human-friendly messages derived from `message` and may implement retry/backoff for transient errors (429, 500, 503). All error responses shall be logged server-side with context for debugging and audit.

3.5 Non-Functional Requirements

Non-functional requirements define the quality attributes that determine how the NEURONET system performs its functions. While functional requirements specify what the system does, non-functional requirements ensure that the system operates efficiently, securely, ethically, and reliably while providing a positive user experience.

3.5.1 Performance Requirements

Performance requirements define responsiveness and efficiency under varying workloads.

- **NFR-01:** The system shall respond to user requests within acceptable response times under normal operating conditions.
- **NFR-02:** The system shall support concurrent access by multiple users without significant performance degradation.
- **NFR-03:** AI-based sentiment analysis shall be processed efficiently to ensure timely availability of insights.
- **NFR-04:** The system shall manage server resources effectively during peak usage periods.
- **NFR-05:** The system shall be scalable to accommodate user growth without major architectural changes.
- **NFR-06:** The system shall respond to interactive user requests within 2 seconds for at least 95% of requests under normal operating conditions.
- **NFR-07:** The system shall support a minimum of 100 concurrent active users during pilot deployment without significant performance degradation.
- **NFR-08:** AI-based sentiment analysis and trend detection results shall be available within 24 hours of data submission under normal operating conditions.
- **NFR-09:** The system shall ensure that alert condition evaluations are completed within a maximum delay of 24 hours for trend-based detection.
- **NFR-10:** The system shall complete standard database read and write operations within 500 milliseconds for at least 95% of transactions under normal load.

3.5.2 Usability Requirements

Usability requirements ensure ease of use and accessibility for all user classes.

- **NFR-11:** The system shall provide an intuitive interface suitable for adolescent users.
- **NFR-12:** The system shall use clear, simple, and supportive language.
- **NFR-13:** The system shall maintain consistent navigation and layout across platforms.
- **NFR-14:** The system shall minimize the steps required to complete common tasks.

- **NFR-15:** The system shall present insights and alerts in a supportive, non-stigmatizing manner.
- **NFR-16:** The system shall achieve an operational availability of at least 99.5% during expected usage periods, excluding scheduled maintenance.

3.5.3 Security Requirements

Security requirements protect against unauthorized access, misuse, and data breaches.

- **NFR-17:** The system shall require authentication before granting access to any functionality.
- **NFR-18:** The system shall enforce strict role-based access control.
- **NFR-19:** All sensitive data shall be encrypted during storage and transmission.
- **NFR-20:** The system shall restrict access to private content such as journals and chats.
- **NFR-21:** The system shall maintain audit logs of critical actions.
- **NFR-22:** Passwords shall be securely stored using hashing and encryption mechanisms.

3.5.4 Software Quality Attributes

Reliability

- The system shall operate consistently with minimal failures.
- The system shall recover gracefully from unexpected errors.

Maintainability

- The system shall follow modular and layered architecture principles.
- The system shall be well documented to support future enhancements.

Scalability

- The system shall support growth in users, data volume, and features.

Portability

- The system shall operate across supported operating systems with minimal modification.

Availability

- The system shall be available during expected usage periods with minimal downtime.

Ethical Compliance

- The system shall respect user privacy and autonomy.

- The system shall clearly communicate its non-clinical purpose and AI limitations.
- The system shall avoid automatic guardian alerts to protect adolescent trust and reduce ethical risk.
- The system shall clearly differentiate between educational content and clinical advice.

3.5.5 Summary of Non-Functional Requirements

The non-functional requirements ensure that the NEURONET system delivers a high-quality, dependable, and user-centered experience while handling sensitive adolescent mental health-related data. By addressing key quality attributes such as performance, usability, security, reliability, and scalability, these requirements define how the system should operate under normal and peak conditions. They guide architectural and technical decisions to ensure fast response times, intuitive interfaces, consistent system behavior, and smooth interaction across both mobile and web platforms for all user roles.

In addition, the non-functional requirements emphasize ethical responsibility, data protection, and long-term maintainability. Strong security controls, role-based access, and encryption mechanisms help maintain user trust and protect privacy, while modular design and documentation support future enhancements and system evolution. Together, these requirements ensure that the NEURONET system not only meets functional expectations but also complies with ethical standards, supports sustainable growth, and provides a safe and supportive digital environment for adolescents, guardians, and counselors.

APPENDIX A

Group Log

This section documents the group activities, decisions, and progress related to the creation of the Software Requirements Specification (SRS) and Software Design Document (SDD) for the NEURONET – AI-Driven Early Detection and Support System for Adolescent Mental Health project.

➤ Meeting 1: December 2, 2025

Agenda:

- Understanding the project scope and objectives for the SRS.
- Allocating roles for the SRS preparation.
- Researching templates and tools for creating the SRS.

Key Decisions:

- Agreed on the functionalities and requirements to be covered in the SRS.
- Roles were assigned to team members for specific tasks related to the SRS, including requirements gathering and drafting sections.
- MS Word for document writing, and Draw.io for diagrams were selected.

Next Plan:

- Start collecting the initial requirements for the SRS.
- Set deadlines for the first draft of the SRS.
- Begin structuring the SRS document based on the provided outline.

➤ Meeting 2: December 13, 2025

Agenda:

- Review the initial draft and progress on the SRS.
- Discuss additional sections to be included in the SRS.
- Address any challenges in documentation and tool usage.

Key Decisions:

- Refined the initial SRS draft and included additional details as needed.
- Discussed the system architecture and user requirements that should be added to the SRS.
- Decided to continue working on the sections regarding system operations and user requirements.

Next Plan:

- Continue drafting and reviewing the SRS sections.
- Prepare for submission of the SRS draft to the advisor for feedback.

➤ Meeting 3: December 20, 2025**Agenda:**

- Finalize the first draft of the SRS.
- Submit the draft to the advisor for feedback.
- Review any incomplete sections of the SRS.

Key Decisions:

- Completed the first draft of the SRS and submitted it to the advisor for review.
- Identified sections in the SRS that required additional detail and refinement.
- Discussed feedback expectations from the advisor and clarified what further improvements were needed.

Next Plan:

- Incorporate advisor feedback to refine the SRS document.
- Review sections that need further detailing or adjustments.

➤ Meeting 4: January 5, 2026**Agenda:**

- Review and finalize the SRS based on advisor feedback.

- Ensure that all sections of the SRS are complete and accurate.
- Prepare the final version of the SRS for submission.

Key Decisions:

- Incorporated all feedback from the advisor to finalize the SRS document.
- Ensured that all required sections, such as functional and non-functional requirements, were included and properly detailed.
- Conducted a final review to check for any missing or unclear sections.

Next Plan:

- Submit the final SRS to the advisor for approval.
- Begin preparation for the Software Design Document (SDD).

➤ Meeting 5: January 10, 2026

Agenda:

- Initiate the Software Design Document (SDD) preparation.
- Allocate roles for SDD sections including architecture, data design, and interface design.
- Review SRS feedback and ensure alignment with SDD.

Key Decisions:

- Started working on the SDD based on the approved SRS.
- Assigned specific sections of the SDD to team members (architecture, class diagrams, subsystem decomposition, data design).
- Agreed on using UML diagrams for system design visualization.

Next Plan:

- Draft initial sections of the SDD.
- Create UML diagrams (use case, class, sequence, deployment).
- Schedule next meeting to review progress.

➤ **Meeting 6: January 14, 2026**

Agenda:

- Review progress on SDD sections.
- Finalize UML diagrams and system architecture design.
- Discuss data management approach and database schema.

Key Decisions:

- Completed major UML diagrams including use case, class, and sequence diagrams.
- Finalized the hybrid database approach (PostgreSQL + MongoDB).
- Reviewed subsystem decomposition and interaction flows.

Next Plan:

- Complete remaining SDD sections (interface design, requirement traceability).
- Prepare for SDD draft submission to advisor.

➤ **Meeting 7: January 16, 2026**

Agenda:

- Finalize the first draft of the SDD.
- Submit the SDD draft to the advisor for feedback.
- Prepare for senior project presentation.

Key Decisions:

- Completed the first draft of the SDD and submitted it to the advisor for review.
- Began preparation for the senior project presentation including PowerPoint slides.
- Assigned presentation sections to team members.

Next Plan:

- Incorporate advisor feedback on SDD.
- Finalize presentation materials.
- Conduct presentation rehearsals.

➤ Advisor Feedback

The team submitted the first draft of the SRS for advisor review. After receiving feedback, adjustments were made to ensure the document met the required standards. The SDD was subsequently developed based on the approved SRS, and feedback was incorporated to finalize both documents.

➤ Progress Reports & Signatures

The progress reports for the SRS and SDD were submitted and reviewed by the advisor. The advisor's signature confirms the completion and approval of the documents.

Team Members

Here are the team members who contributed to the Software Requirements Specification (SRS) and Software Design Document (SDD) for the NEURONET – AI-Driven Early Detection and Support System for Adolescent Mental Health project:

1. Garoma Makure Gebeyo (ID: 2992/14) – Team Lead and Developer



2. Hamdi Abdulfetah (ID: 3017/14) – Frontend Developer



3. Firafis Berhanu Gemechu (ID: 2937/14) – Backend Developer



4. Muaz Shewabu Redi (ID: 3543/14) – UI/UX Designer

